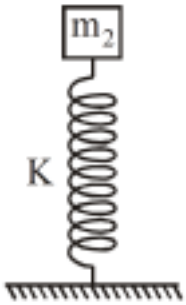
PHYSICS

1) Block of mass m_2 is in equilibrium as shown in figure. Another block of mass m_1 is kept gently on m_2. Find the amplitude of oscillation:-



(1) $\frac{m_1 g}{K}$

(2) $\frac{2m_1 g}{K}$

(3) $\frac{3m_1 g}{K}$

(4) $\frac{m_1 g}{2K}$

2) Which of the following is never positive in a SHM
[$\vec{F}$: → restoring force, $\vec{a}$: → acceleration, $\vec{r}$: → displacement, $\vec{v}$: → velocity] ?

(1) $\vec{F} \cdot \vec{a}$

(2) $\vec{v} \cdot \vec{r}$

(3) $\vec{F} \cdot \vec{r}$

(4) $\vec{F}.\vec{v}$

3) Time period of a spring-mass system can be changed by:-
(A) Changing the mass
(B) Accelerating the point of suspension of the mass
(C) Cutting the spring (i.e. changing the length of the spring)
(D) Immersing the mass in a liquid

(1) A, B

(2) A, C

(3) B, C

(4) D, B

4) A weightless spring of length 60 cm and force constant 100 N/m is kept straight and unstretched on a smooth horizontal table and its ends are rigidly fixed. A small mass of 0.25 kg is attached firmly at the middle of the spring and is slightly displaced along the length and released. The time period of the oscillation of the mass is :-

(1) $\frac{\pi}{20}$ sec

(2) $\frac{\pi}{10}$ sec

(3) $\frac{\pi}{5}$ sec

(4) $\frac{\pi}{\sqrt{200}}$ sec

5) **Statement I** :- When a particle is at extreme position performing SHM, its momentum is zero.
Statement II : At extreme position, the velocity of particle, performing SHM is zero.
Choose the correct option :-

(1) Both statements are true

(2) Statement I is false but statement II is true

(3) Statement I is true but statement II is false

(4) Both the statements are false.

6) A wire of length '□' having tension T and radius 'r' vibrates with fundamental frequency 'f'. Another wire of the same metal with length 2□ having tension 2T and radius 2r will vibrate with fundamental frequency :-

(1) f

(2) 2f

(3) $\frac{f}{2\sqrt{2}}$

(4) $\frac{f}{2}\sqrt{2}$

7) A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is:-

(1) 330 m/s

(2) 339 m/s

(3) 350 m/s

(4) 300 m/s

8) The equation of a wave on a string of linear mass density 0.04 kg m^{-1} is given by
$y = 0.02(m) \sin\left[2\pi\left(\frac{t}{0.02(s)} - \frac{x}{0.50(m)}\right)\right]$.
The tension in the string is :

(1) 6.25 N

(2) 12.5 N

(3) 20 N

(4) 25 N

9) The mathematical forms for three sinusoidal traveling waves are given by
Wave 1 : y(x,t) = (2cm) sin(3x–6t)
Wave 2 : y(x,t) = (3cm) sin(4x–12t)
Wave 3 : y(x,t) = (4cm) sin(5x–11t)
where x is in meters and t is in seconds. Of these waves :

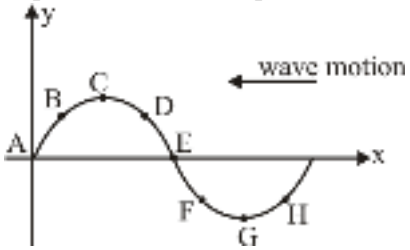
(1) wave 1 has the greatest wave speed and the greatest maximum transverse string speed.

(2) wave 2 has the greatest wave speed and wave 1 has the greatest maximum transverse string speed.

(3) wave 3 has the greatest wave speed and the greatest maximum transverse string speed.

(4) wave 2 has the greatest wave speed and wave 3 has the greatest maximum transverse string speed.

10) A transverse wave is travelling along a stretched string from right to left. The figure shown represents the shape of the string (snap short) at a given instant. At this instant



(1) the particles at A, B and H have downward velocity

(2) the particles at D, E and F have downward velocity

(3) the particles at C, E and G have zero velocity

(4) the particles at A and F have maximum velocity

11) A tuning fork resonates with a sonometer wire of length 1 m stretched with a tension of 6 N. When the tension in the wire is changed to 54 N, the same tuning fork produces 12 beats per second with it. The frequency of the tuning fork is _____ Hz.

(1) 2

(2) 1

(3) 6

(4) 3

12) A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100 gm, filled with 170 gm of water at room temperature. Subsequently, temperature of the system is found to be 75°C. Then T is given by :
Given room temperature = 30°C
Specific heat of copper = 0.1 cal/gm°C

(1) 1250°C

(2) 825°C

(3) 800°C

(4) 885°C

13) Select the correct graphs :

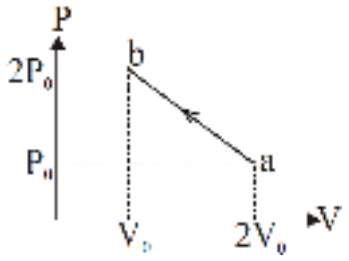
(i) P – $\frac{1}{V}$ graph at constant temperature is a rectangular hyperbola
(ii) PV – V graph is a straight line parallel to V-axis at const. temperature
(iii) P – V graph at constant temperature is a straight line passing through origin
(iv) V – T graph at constant pressure is a straight line passing through origin

(1) i, iii
(2) i, iv
(3) ii, iii
(4) ii, iv

14) 200 gm water is filled in a calorimetry of negligible heat capacity. It is heated till its temperature is increase by 20ºC. Find the heat supplied to the water.

(1) 1000 cal
(2) 2000 cal
(3) 3000 cal
(4) 4000 cal

15) One mole of monoatomic ideal gas undergoes the thermodynamic process a→b shown in the P–V diagram. During this process maximum temperature will occur when gas volume will be :



(1) V_0
(2) $2V_0$
(3) $\frac{3}{2}V^0$
(4) $\frac{5}{4}V^0$

16) A thin walled metal tank of surface area $5m^2$ is filled with water and contains an immersion heater dissipating 1kW. The tank is covered with 4 cm thick layer of insulation whose thermal conductivity is 0.2 watt/ m-K. The outer face of the insulation is at 25°C. The temperature of the tank in the steady state is -

(1) –15°C
(2) 65°C
(3) 20°C
(4) 90°C

17) A container has two chambers of volumes V_1 = 2 litres and V_2 = 3 litres separated by a partition made of a thermal insulator. The chambers contains n_1 = 5 and n_2 = 4 moles of ideal gas at

pressures p_1 = 1 atm and p_2 =2 atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of :

(1) 1.3 atm

(2) 1.6 atm

(3) 1.4 atm

(4) 1.8 atm

18) In an adiabatic process, where pressure is increased by 6%. If $\frac{C_P}{C_V} = \frac{3}{2}$, than the volume decreases by about :

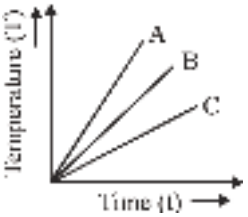
(1) 4%

(2) 2%

(3) $\frac{3}{2}$%

(4) $\frac{2}{3}$%

19) Which of the substances A, B or C of same mass has the highest specific heat if heat supply rate is same? The temperature v/s time graph is shown



(1) A

(2) B

(3) C

(4) All have equal specific heat

20) The rate radiation emitted by a star A is 10,000 times that of the star B. If the surface temperatures of the star B and star A are 6000 K and 2000 K respectively, the ratio of the radii of the star A and star B is :-

(1) 300 : 1

(2) 600 : 1

(3) 900 : 1

(4) 1200 : 1

21) At what temperature, rms velocity of O_2 molecules will be 1/3 that of H_2 molecules at –3°C?

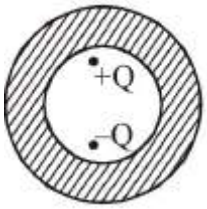
(1) 90 K

(2) 1167°C

(3) –3°C

(4) 207°C

22) Shown in the figure are two point charges +Q and − Q inside the cavity of a spherical shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ is



surface charge density on outer surface and q is net charge on outer surface then,

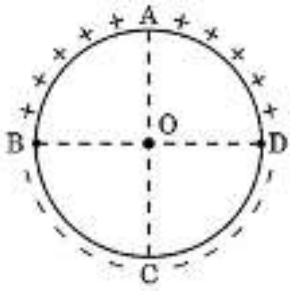
(1) Only $\sigma = 0$

(2) Only $q = 0$

(3) Both σ and q are zero

(4) None of the above

23) **Assertion (A) :** Equal amount of positive and negative charges are distributed uniformly on two halves of a thin circular ring as shown in figure. The resultant electric field at the centre O of the ring is along OC.



Reason (R) : It is so because the net potential at O is not zero.

(1) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) is false and Reason (R) is also false.

24) Two point charges $q_1 = 20\mu C$ and $q_2 = 25\mu C$ are placed at (– 1, 1,1) m and (3,1, – 2)m, with respect to a coordinate system. Unit vector along electrostatic force on q_2 ?

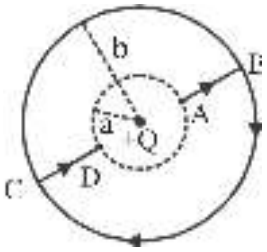
(1) $\frac{4\hat{i}-3\hat{k}}{5}$

(2) $\frac{2\hat{i}-3\hat{k}}{5}$

(3) $4\hat{i}-3\hat{k}$

(4) $2\hat{j}-3\hat{k}$

25) Two concentric circles are drawn with a charge +Q at their center. Now a small test charge q is taken from point A to point D along the path ABCD. If the radii of the circles are a and b, then the



amount of work done is :-

(1) $kqQ\left(\frac{1}{a}-\frac{1}{b}\right)$

(2) $kqQ\left(\frac{1}{a}+\frac{1}{b}\right)$

(3) zero

(4) none of the above

26) Concentric metallic hollow spheres of radii R and 4R hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference V(R) – V(4R) is :

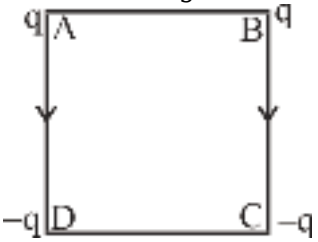
(1) $\frac{3Q_1}{16\pi\varepsilon_0 R}$

(2) $\frac{Q_2}{4\pi\varepsilon_0 R}$

(3) $\frac{3Q_1}{4\pi\varepsilon_0 R}$

(4) $\frac{3Q_2}{4\pi\varepsilon_0 R}$

27) Charges are placed on the vertices of a square as shown. Let E be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C



respectively, then :-

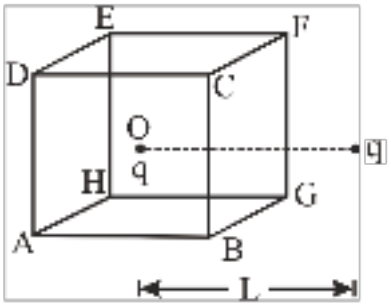
(1) $\vec{E}$ remains unchanged, V changes

(2) Both $\vec{E}$ and V change

(3) $\vec{E}$ and V remains unchanged

(4) $\vec{E}$ changes, V remains unchanged

28) A charged particle q is placed at the centre O of cube of length L (ABCDEFGH). Another same charge q is placed at a distance L from O. Then, the net electric flux through BCFG is :-



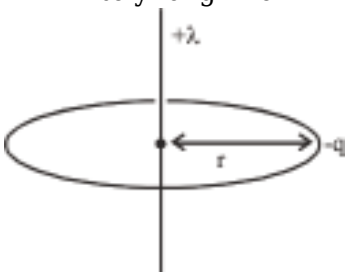
(1) $\frac{q}{4\pi\varepsilon_0 L}$

(2) Zero

(3) $\frac{q}{2\pi\varepsilon_0 L}$

(4) $\frac{q}{3\pi\varepsilon_0 L}$

29) A particle of charge –q & mass *m* moves in a circle of radius r around an infinitely long line charge of charge density +λ. Then time period will be (Where k = $\frac{1}{4\pi\in_0}$)



(1) $T = 2\pi\, r\sqrt{\frac{m}{2k\lambda q}}$

(2) $T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$

(3) $T = \frac{1}{2\pi r}\sqrt{\frac{2k\lambda q}{m}}$

(4) $T = \frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$

30) ***Assertion*** :- When a capacitor is charged by a battery, both the plates receive charges equal in magnitude, no matter whether the sizes of plates are identical or not. ***Reason*** :- The charge distribution on the plates of a capacitor is in accordance with the charge conservation principle.

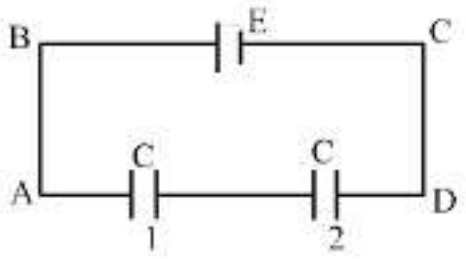
(1) Both Assertion & Reason are true & the reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are true but reason is not a correct explanation of the Assertion.

(3) Assertion is true but the Reason is false

(4) Both Assertion & Reason are false

31) In the adjoining figure, capacitor (1) and (2) have a capacitance 'C' each. When the dielectric of dielectric constant K is inserted between the plates of one of the capacitor, the total charge flowing



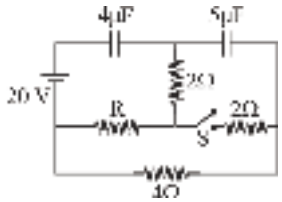
through battery is :-

(1) $\frac{KCE}{K+1}$ from B to C

(2) $\frac{KCE}{K+1}$ from C to B

(3) $\frac{(K-1)CE}{2(K+1)}$ from B to C

(4) $\frac{(K-1)CE}{2(K+1)}$ from C to B



32) Find heat produced in the capacitors on closing the switch S :-

(1) 0.0002 J

(2) .0005 J

(3) 0.00075 J

(4) zero

33) A conducting sphere of radius 10 cm is charged by 10 μC. Another uncharged sphere of radius 20 cm is allowed to touch it for some time. After that if the sphere are separated, then surface density of charges, on the spheres will be in the ratio of :–

(1) 1 : 4

(2) 1 : 3

(3) 2 : 1

(4) 1 : 1

34) 12 identical cells are connected in series with some cells wrongly connected. The arrangement is connected in series with an ammeter and two such more cells in series forming a battery. Current is 3 A when cells and battery aid each other and is 2 A when cells and battery oppose each other. The number of cells wrongly connected is

(1) 4

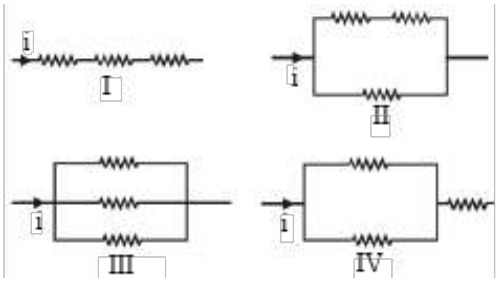
(2) 1

(3) 3

(4) 2

35) Three resistances of equal value are arranged in different combinations as shown below.

Arrange them in the increasing order of power dissipation :



(1) III < II < IV < I

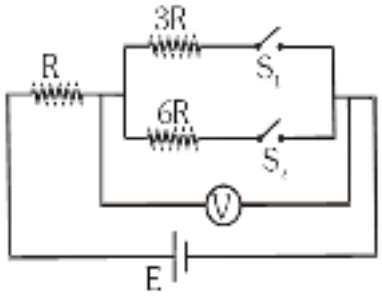
(2) II < III < IV < I

(3) I < IV < III < II

(4) I < III < II < IV

36) In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed and reading of voltmeter is V_3 when both S_1 and S_2 are closed.

Then :-



(1) $V_3 > V_2 > V_1$

(2) $V_2 > V_1 > V_3$

(3) $V_3 > V_1 > V_2$

(4) $V_1 > V_2 > V_3$

37) Two conductors have same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2. The equivalent temperature coefficients of their series combinations are nearly:-

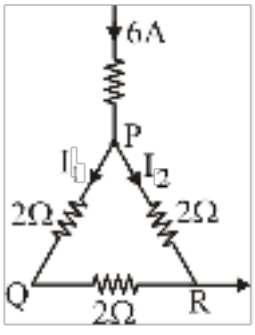
(1) $\frac{\alpha_1 + \alpha_2}{2}$

(2) $\alpha_1 + \alpha_2$

(3) $2(\alpha_1 + \alpha_2)$

(4) $\frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$

38) A current of 6A enters one corner P of an equilateral triangle PQR having 3 wires of resistances 2Ω each and leaves by the corner R. Then, the current I_1 and I_2 are:-



(1) 2A, 4A

(2) 4A, 2A

(3) 1A, 2A

(4) 2A, 3A

39) A charged particle having drift velocity of 7.5×10^{-4} ms^{-1} in an electric field of 3×10^{-10} Vm^{-1}, has a mobility in m^2V^{-1} s^{-1} of:

(1) 2.25×10^{-15}

(2) 2.25×10^{15}

(3) 2.5×10^{6}

(4) 2.5×10^{-6}

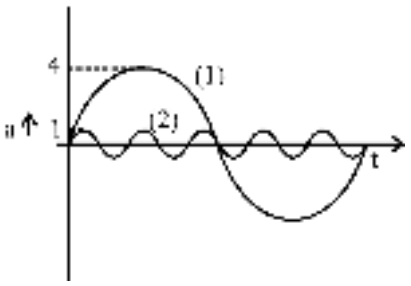
40) A galvanometer of coil resistance 45 Ω gives full deflection at 1A current. Value of shunt resistance required to convert it in 10 A range ammeter is :

(1) 4.5 Ω

(2) 9 Ω

(3) 5 Ω

(4) None of these



41) For a given medium find intensity ratio $I_1 : I_2$.

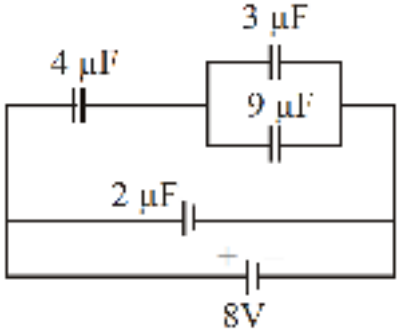
(1) 4 : 5

(2) 5 : 4

(3) 16 : 25

(4) 25 :16

42) A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the 4 µF and 9 µF capacitors), at a point 30 m from it , would equal:



(1) 480 N/C

(2) 240 N/C

(3) 360 N/C

(4) 420 N/C

43) The net resistance of circuit between A and B is:- 

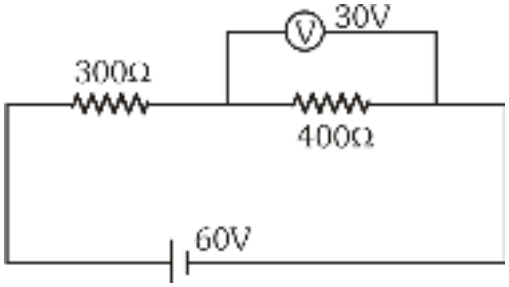
(1) 10 Ω

(2) 12 Ω

(3) 8 Ω

(4) 16 Ω

44) In the circuit shown here, the voltmeter reads 30V. What is the resistance of the voltmeter –



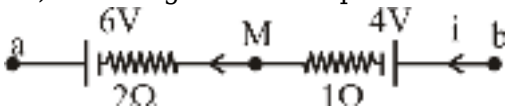
(1) 1200 Ω

(2) 700 Ω

(3) 400 Ω

(4) 300 Ω

45) For the given circuit potential at a and b is 10V and 14V then potential of point M is :-



(1) 8V

(2) 7V

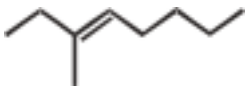
(3) 6V

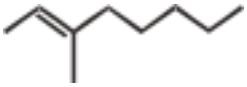
(4) 5V

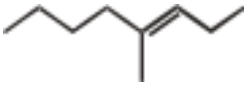
CHEMISTRY

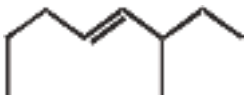
1) Alkene (X) 

X is :-

(1) 

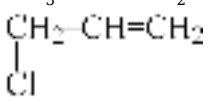
(2) 

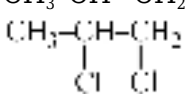
(3) 

(4) 

2) Which of the following reactions is incorrect ?

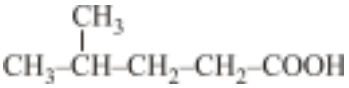
(1) $CH_3CH_2COONa \xrightarrow{\text{Electrolysis}} CH_3CH_2CH_2CH_3$

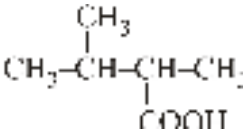
(2) $CH_3\text{–}CH{=}CH_2 \xrightarrow{Cl_2/\Delta}$ 

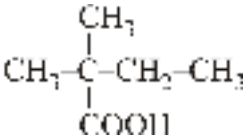
(3) $CH_3\text{–}CH{=}CH_2 \xrightarrow{Cl_2/H_2O}$ 

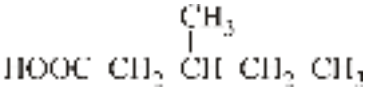
(4) $CH_3CH_2Br \xrightarrow{\text{Na/Dry ether}} CH_3CH_2CH_2CH_3$

3) Which of the following acid undergoes fastest decarboxylation ?

(1) 

(2) 

(3) 

(4) 

4) Which of the following is correct order of basic strength in aqueous phase ?

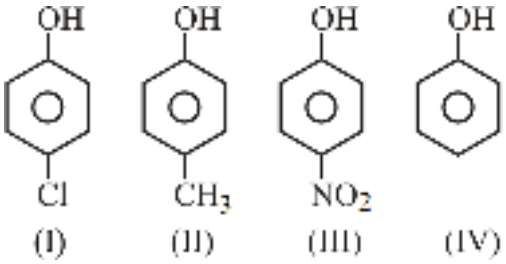
(1) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$

(2) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH > NH_3$

(3) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(4) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2 > NH_3$

5) Arrange the following compound in the correct order of pK_a ?



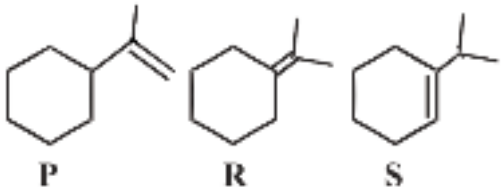
(1) III > I > IV > II

(2) III > IV > I > II

(3) II > IV > I > III

(4) III > II > IV > I

6) Arrange the following alkenes in increasing order of their heat of hydrogenation :-



(1) R < S < P

(2) R = S = P

(3) P < R < S

(4) P < S < R

7) The correctly matched pairs of the following are given by the sets :
(A) Neopentane – position isomer of isopentane
(B) Isopentane – three primary carbon atoms
(C) Isobutylene – no secondary carbon
(D) Isobutane – two primary carbons

(1) A and B

(2) B and C

(3) C and D

(4) A and D

8) Which of the following compound can be named by taking iso as a prefix ? [Common Name]

(1) 2,2,3,3-Tetramethyl pentane

(2) 2,2-Dimethyl pentane

(3) 2,2,3-Trimethyl pentane

(4) 2-methyl pentane

9)

Match column-I with column-II.

	Column-I (Reaction)		Column-II (Intermediate)
(A)	$CH_3–CH_2–CH_3 \xrightarrow[h\nu]{Br_2}$	(P)	$\dot{C}H_2–CH=CH_2$
(B)	$CH_3–CH=CH_2 \xrightarrow{HBr}$	(Q)	$CH_3–\dot{C}H–CH_3$
(C)	$CH_3–CH=CH_2 \xrightarrow[R_2O_2]{HBr}$	(R)	$CH_3–\overset{\oplus}{C}H–CH_3$
(D)	$CH_3–CH=CH_2 \xrightarrow{NBS}$	(S)	$CH_3\ \dot{C}H\ CH_2$ (Br)

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-R, C-S, D-P

(3) A-S, B-R, C-Q, D-P

(4) A-Q, B-S, C-R, D-P

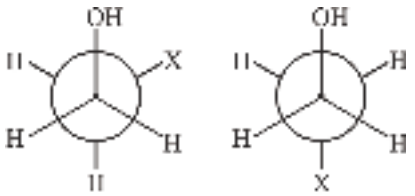
10) The correct statement for Isooctane is –

(1) It has seven carbons in its principal carbon chain.

(2) It has five secondary carbons.

(3) It has one quarternary carbon and one tertiary carbon.

(4) It has ten primary hydrogens.



11) (A) (B)

A will be more stable than B if X is :

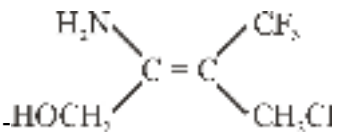
(1) $–CH_3$

(2) $–C_2H_5$

(3) –F

(4) –I

12) Configuration of given compound is :-



(1) cis

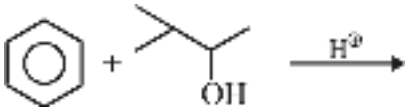
(2) trans

(3) E

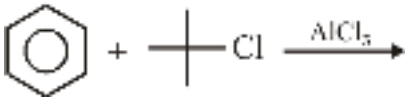
(4) Z

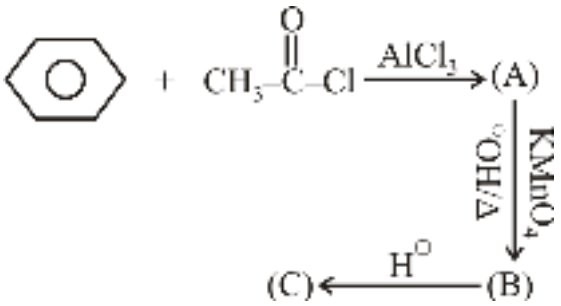
13) Which of the following reaction give secondary butyl benzene as product ?

(1) 

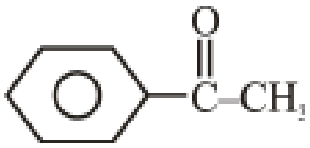
(2) 

(3) 

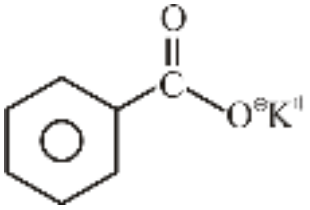
(4) 

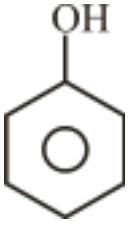
14) 

Which of the following product is not formed in given reaction sequence.

(1) 

(2) 

(3) 

(4) 

15) Give the correct order of increasing acidity of the following compounds :-

$Cl–CH_2–COOH$ (a) $CH_3–CH_2COOH$ (b)

$Cl–CH_2–CH_2COOH$ (c) $(CH_3)_2CHCOOH$ (d)

CH_3COOH

(e)

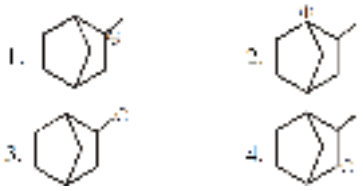
(1) e < b < d < a < c

(2) d < b < e < c < a

(3) c < a < d < b < e

(4) e < d < b < a < c

16) Find out correct stability order in the following carbocations :-



(1) 4 > 1 > 3 > 2

(2) 1 > 4 > 3 > 2

(3) 1 > 2 > 4 > 3

(4) 2 > 1 > 4 > 3

17) **Assertion (A) :** If an organic compound contain nitrogen or sulphur then test for halogen using sodium fusion extract may gives incorrect result.
Reason (R) : During sodium fusion nitrogen and sulphur form $NaCN/Na_2S/NaSCN$ which interfere with $AgNO_3$ test for halogen.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

18) Distillation under reduced pressure is used to purify liquids which?

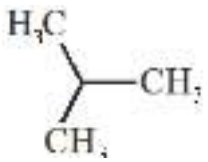
(1) Are highly volatile

(2) Have high boiling points & decomposes at or below their boiling point

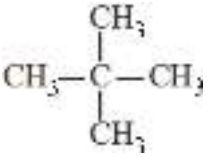
(3) Are explosive

(4) Have low boiling points

19) Which of the following alkane give alcohol in presence of $KMnO_4$.

(1) 

(2) $CH_3–CH_2–CH_3$

(3) 

(4) CH_4

20) Which of the following will give only three monochloro structural product ?
(a) $CH_3CH_2CH_2CH_2CH_2CH_3$
(b) $CH_3–CH_2–C(CH_3)_2CH_3$
(c) $CH_3CH_2CH_2CH(CH_3)CH_3$
(d) $CH_3CH(CH_3)CH(CH_3)CH_3$

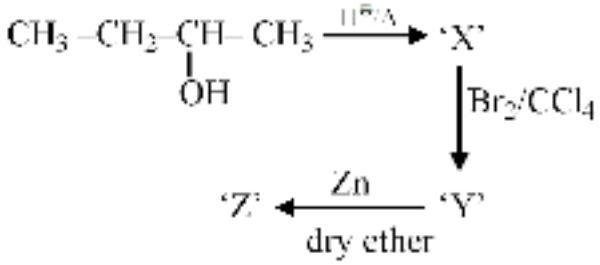
(1) a, b

(2) b, c

(3) a, c

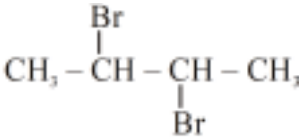
(4) b, d

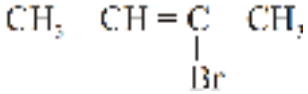
21)



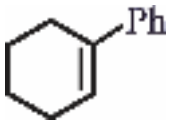
'Z' is :-

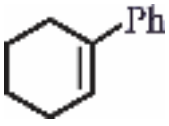
(1) $CH_3 – CH = CH – CH_3$

(2) 

(3) 

(4) $H_2C = CH – CH = CH_2$

22)  + HBr → X

 + HBr $\xrightarrow{H_2O_2}$ Y,

Relationship between X and Y are.

(1) Position isomers

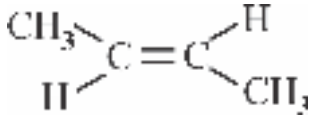
(2) Enantiomers

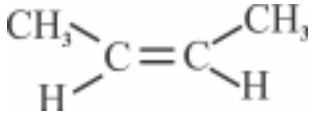
(3) Chain isomers

(4) Geometrical isomers

23) In the following reaction product (Z) is

$CH_3–CH=CH–CH_3 \xrightarrow[CCl_4]{Br_2} X \xrightarrow[(2)\ NaNH_2,\Delta]{(1)\ Alc.KOH} Y \xrightarrow[\Delta]{Na+Liq.NH_3} Z$

(1) 

(2) 

(3) $CH_3–CH_2–CH_2–CH_3$

(4) $CH_3 - C \equiv C - CH_3$

24)

Which of the following statement is incorrect?

(1) Rate of reaction always depends on concentration of reactant

(2) Catalyst can either increase or decrease the rate of reaction

(3) Rate of reaction changes with temperature

(4) For solid reactant, larger the surface area, faster will be the reaction

25) In a first order reaction, the concentration of the reactant, 30 minutes and 40 minutes after the start are C_1 and C_2 (in moles/litre) respectively. What was C_0, initial concentration?

(1) $C_0 = \left[\frac{C_1^3}{C_2^4}\right]$

(2) $C_0 = \left[\frac{C_1}{C_2}\right]^4$

(3) $C_0 = \left[\frac{C_1}{C_2}\right]^3$

(4) $C_0 = \left[\frac{C_1^4}{C_2^3}\right]$

26) **Assertion :** In Maxwell Boltzmann distribution curve, the fraction of molecules with most probable kinetic energy increases at higher temperature.
Reason : Most probable kinetic energy of molecules increases with increase in temperature.
Choose the correct option :

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

27) Which of the following statement is incorrect ?

(1) $t_{1/2}$ (half life) increases with increase in temperature.

(2) Zero order reaction is a complex reaction.

(3) Rate constant increases with increase in temperature.

(4) Activation energy remains same with increase in temperature.

28) In the following reaction, $2H_2O_2 \rightarrow 2H_2O + O_2$ rate of formation of O_2 is 36 gL^{-1} min^{-1}, what is rate of disappearance of H_2O_2 :

(1) 2.250 mol L^{-1} min^{-1}

(2) 72 mol L^{-1} min^{-1}

(3) 1.50 mol L^{-1} min^{-1}

(4) 38.25 mol L^{-1} min^{-1}

29) On increasing temperature from 200K to 220K, rate of reaction A increases by 3 times and rate of reaction B increases by 9 times then correct relationship between activation energy of A and B is :-

(1) $E_A = 3E_B$

(2) $3E_A = E_B$

(3) $E_B = 2E_A$

(4) $E_A = 2E_B$

30) Match the following Column-I with Column-II.

	Column-I		**Column-II**
(A)	Raoult's law	(P)	$\Delta T_f = K_f m$
(B)	Henry's law	(Q)	$\pi = CST$
(C)	Elevation of boiling point	(R)	$p = x_1 p_1^o + x_2 p_2^o$
(D)	Depression in freezing point	(S)	$\Delta T_b = K_b m$
(E)	Osmotic pressure	(T)	$p = K_H x$

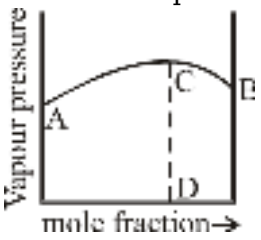
(1) (A)–(S); (B)–(R); (C)–(Q); (D)–(P); (E)–(T)

(2) (A)–(R); (B)–(T); (C)–(S); (D)–(P); (E)–(Q)

(3) (A)–(T); (B)–(R); (C)–(Q); (D)–(P); (E)–(S)

(4) (A)–(S); (B)–(R); (C)–(T); (D)–(P); (E)–(Q)

31) The diagram given below is a vapour pressure composition diagram for a binary solution of A and B. In the solution, A-B interaction are



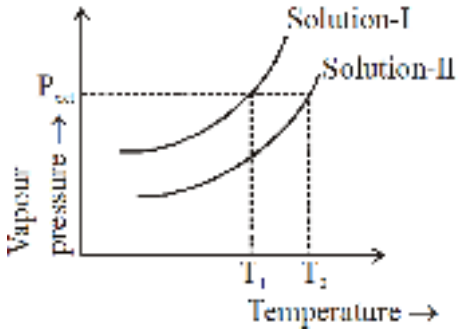
(1) Similar to A-A and B-B interactions

(2) Greater than A-A and B-B interactions

(3) Smaller than A-A and B-B interactions

(4) Unpredictable

32) Two different solutions were prepared by dissolving sucrose in water having concentration of 1m and 2m respectively. Value of $(T_2–T_1)$ will be:-



[K_b = 0.52 K Kg. m^{-1}]

(1) 1 K

(2) 1.86 K

(3) 0.52 K

(4) 1.86° C

33) ***Assertion (A)*** :- RBC when placed in pure water will swell up and burst.
Reason (R) :- When RBC is placed in pure water endo osmosis will take place and pressure inside RBC will increase.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

34) How many grams of carbon dioxide gas is dissolved in a 1L bottle of carbonated water,if the manufacturer uses a pressure of 2.4 atm in the bottling process at 25°C ? (K_H of CO_2 = 29.76 $\frac{atm}{M}$) :-

(1) 8.06 g

(2) 3.52 g

(3) 4.59 g

(4) 6.91 g

35) Two liquids A and B form ideal solution at 300 K. The vapour pressure of a solution containing one mole of A and four mole of B is 560 mm of Hg. At the same temperature if one mole of B is taken out from the solution, the vapour pressure of solution decreases by 10 mm of Hg. Vapour pressures of A and B in pure state will be-

(1) 330, 550

(2) 200, 300

(3) 400, 600

(4) 800, 1200

36) The freezing point depression constant for water is +1.86°C m^{-1}. If 5.00 g Na_2SO_4 is dissolved

in 50.0 g H_2O, the freezing point is changed by –3.72°C. Calculate the van't Hoff factor for Na_2SO_4

(1) 2.05
(2) 2.84
(3) 3.11
(4) 0.381

37) 200 cm^3 of 0.1M H_2SO_4 solution is mixed with 150 cm^3 of 0.2M KOH solution. Find the value of evolved heat. [Enthalpy of neutralization is –57.2 kJ/equivalent]

(1) 1.7 kJ
(2) 2.7 kJ
(3) 9.2 kJ
(4) 3.2 kJ

38) The heat of solution of anhydrous $CuSO_4$ and $CuSO_4.5H_2O$ are –15.89 and 2.80 kcal/mol respectively. What will be heat of hydration of anhydrous $CuSO_4$?

(1) –18.69 kcal
(2) 18.69 kcal
(3) –28.96 kcal
(4) 28.96 kcal

39) Which of the following statement is correct ?

(1) Enthalpy of solution for anhydrous copper sulphate and hydrated copper sulphate is same.
(2) Solubility of all compounds in water always increases on increasing temperature.
(3) Litmus can be used as indicator in the permanganate titration.
(4) $KMnO_4$ solution is not regarded as primary standard.

40)

Match the following Column - I and Column -II -

Column - I (Electrolyte)		**Column - II (Product obtained at anode and cathode respectively)**	
(a)	Aqueous NaCl	(p)	$S_2O_8^{-2}$ and H_2
(b)	Fused NaCl	(q)	O_2 and H_2
(c)	Dilute H_2SO_4	(r)	Cl_2 and H_2
(d)	Concentrated H_2SO_4	(s)	Cl_2 and Na

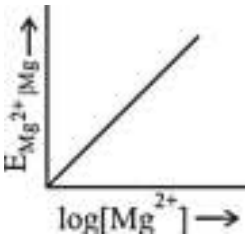
(1) (a) - (q); (b) - (r); (c) - (s); (d) - (p)
(2) (a) - (r); (b) - (s); (c) - (q); (d) - (p)

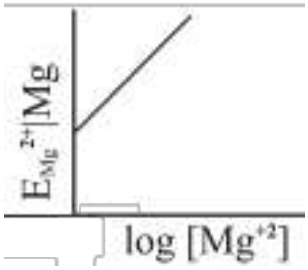
(3) (a) - (q); (b) - (s); (c) - (p); (d) - (r)

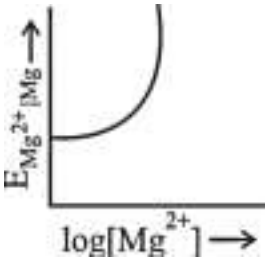
(4) (a) - (r); (b) - (q); (c) - (s); (d) - (p)

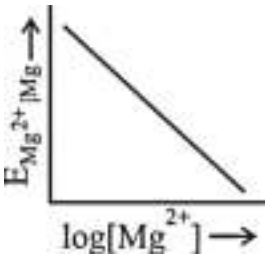
41) Electrode potential for Mg electrode varies according to the equation

$E_{Mg^{2+}|Mg} = E^{o}_{Mg^{2+}|Mg} - \frac{0.059}{2} \log \frac{1}{[Mg^{2+}]}$ The graph of $E_{Mg^{2+}|Mg}$ vs $\log [Mg^{2+}]$ is

(1) 

(2) 

(3) 

(4) 

42) λ_{CH_3COOH} = 20 ohm^{-1} cm^2 eq^{-1} and
$\lambda^{\infty}_{CH_3COOH}$ = 400 ohm^{-1} cm^2 eq^{-1}
calculate the pH of 1M solution of this acid :-

(1) 1.3

(2) 0

(3) 1.7

(4) 4

43) The resistance of 0.1 N solution of formic acid is 200 ohm and cell constant is 2.0 cm^{-1}. The equivalent conductivity (in Scm^2eq^{-1}) of 0.1 N formic acid is :-

(1) 100

(2) 10

(3) 1

(4) None of these

44) The current required to produce oxygen at the rate of 2ml per second at NTP during electrolysis of acidulated water is :-

(1) 8.6 amps/sec

(2) 17.23 amps/sec

(3) 72.6 amps/sec

(4) 34.48 amps/sec

45) **Assertion (A) :** Gold chloride ($AuCl_3$) solution cannot be stored in a vessel made of Copper, iron, nickel, chromium, zinc.
Reason (R) : Gold has very high standard oxidation potential.

(1) Both assertion and reason are true but reason is not correct explanation of assertion

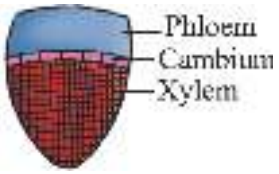
(2) Assertion is true but reason is false

(3) Assertion is false but reason is true

(4) Both assertion and reason are true and reason is a correct explanation of assertion

BIOLOGY

1) Select the correct option in respect of given diagram:-



(1) Present in dicot & gymnosperm stems.

(2) Present in monocot stems

(3) Present in dicot roots

(4) Present in fern rhizome

2) Select the correct option regarding the following statement -
a. Polyarch condition
b. Pith is large and well developed
c. Do not undergo any secondary growth.

(1) Dicot root

(2) Monocot root

(3) Dicot stem

(4) Monocot stem

3) **Assertion:** In dicot stem, endodermis is also referred to as the starch sheath.
Reason: The cells of endodermis are rich in starch.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

4) Select the **incorrect** statement with respect to gymnosperm :-

(1) Have albuminous cells in place of companion cells

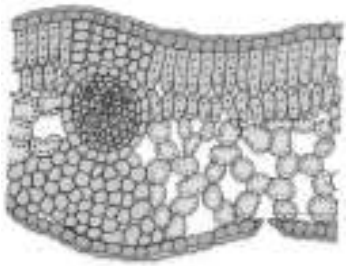
(2) Have sieve tubes at the place of sieve cells

(3) Vessels are generally absent

(4) Tracheids are generally present

5)

Observe the given diagram carefully and select the options which includes all the correct ones.



(A) The position of xylem in the vascular bundle is adaxial.
(B) The positon of *phloem* in the vascular bundle is abaxial.
(C) Adaxial epidermis have certain Bulliform cells (D) The adaxially placed palisade parenchyma is made up of elongated cells which are arranged vertically and parallel to each other. **Options :-**

(1) Statements A, B and C

(2) Statements A, B and D

(3) Only statements A and C

(4) Statements A, C and D

6)  Above type of phyllotaxy can be seen in :-

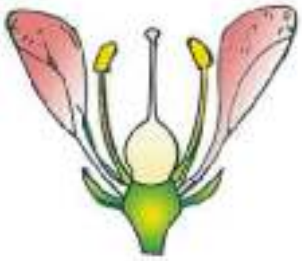
(1) Guava

(2) Chinarose

(3) *Alstonia*

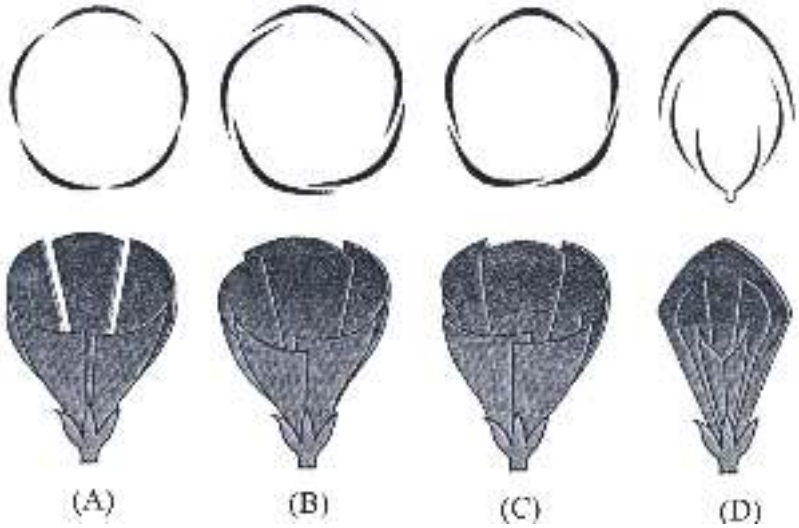
(4) Mustard

7) According to given diagram, find out the type of flower and position of ovary :-



(1) Epigynous - ovary superior
(2) Hypogynous - ovary inferior
(3) Hypogynous - ovary superior
(4) Epigynous - ovary inferior

8) Match the following aestivation with related plants ?



(i) *Cassia*, gulmohur (ii) Pea, bean
(iii) *Calotropis* (iv) China rose

(1) A - (iv), B - (i), C - (ii), D - (iii)
(2) A - (iii), B - (iv), C - (ii), D - (i)
(3) A - (iii), B - (iv), C - (i), D - (ii)
(4) A - (iv), B - (iii), C - (i), D - (ii)

9) Single shield shaped cotyledon of maize seed is called :-

(1) Coleorrhiza
(2) Coleoptile
(3) Scutellum
(4) Epiblast

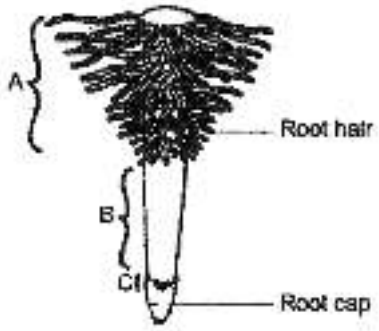
10) Characters of Malvaceae are as following.

(1) Monoadelphous, monothecous and axile placentation.
(2) Dithecous stamen, monoadelphous stamen and unilocular ovary

(3) 5-stigma, stamen extrose and inferior ovary.

(4) 5-Stamen, Diadelphous stamen and apocarpous ovary

11) The below figure is related to the root-tip Identify zones A,B and C.



(1) A-zone of elongation, B-zone of meiosis, C-zone of mitosis

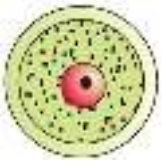
(2) A-zone of maturation, B-zone of meristematic activity C-zone of elongation

(3) A-zone of mitosis, B-zone of elongation C-zone of root cap

(4) A-region of maturation B-region of elongation C-meristematic activity

12) Few diagram regarding the development of pollen grain are given below, arrange these diagrams in correct order and find out the correct option from options given below :-

(A) 

(B) 

(C) 

(D) 

(E) 

(1) A, B, C, D, E

(2) A, B, D, C, E

(3) A, C, D, B, E

(4) A, C, B, D, E

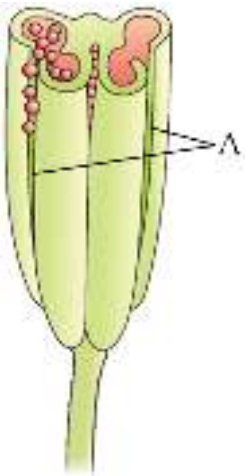
13)  Identify the diagram :-

(1) *Hibiscus*, multicarpellary, syncarpous Gynoecium

(2) Multicarpellary, syncarpous pistil of *Papaver*

(3) Multicarpellary, apocarpous Gynoecium of *Michelia*.

(4) Multicarpellary, syncarpous gynoecium of *Michelia*



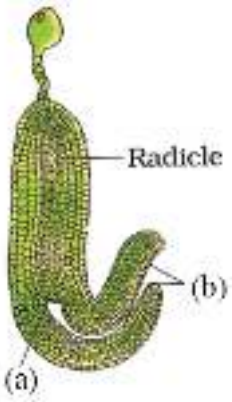
14) Identify A in the given diagram-

(1) Filament

(2) Anther

(3) Line of Dehiscence

(4) Pollen grains



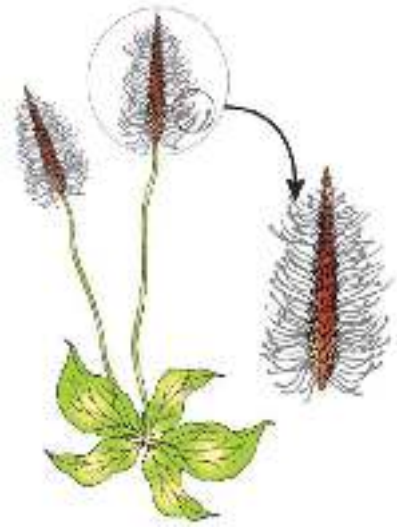
15)
Identify (a) and (b) in given diagram.

(1) (a) Hypocotyl (b) root cap.

(2) (a) Plumule (b) Endosperm

(3) (a) Coleoptile (b) Cotyledons

(4) (a) Plumule (b) Cotyledons

16)  Which statement is wrong for this figure ?

(1) In this plant compact inflorescence is present with well exposed stamens.

(2) It possesses large often feathery stigma.

(3) If often possesses single ovule in each ovary having flowers.

(4) Pollination in such flowers in these plants is less common amongst abiotic pollinators.

17) Which of the following statements are **true**?
(i) Main plant body of angiosperm is a gametophyte.
(ii) Flower is a modified shoot.
(iii) Nodes and Internodes are absent on thalamus.
(iv) Wheat is an annual and monocarpic plant.

(1) i, iii

(2) Only ii

(3) ii, iv

(4) i, ii, iv

18) Select the correct statement.

(1) Exine – Thick, hard, continuous and made of sporopollenin

(2) Germ Pore – 1 or 3, covered by sporopollenin

(3) Pollen grain – generally spherical & 25-50 mm in diameter

(4) The exine exhibits a fascinating array of patterns and designs

19) Select the incorrect matched pair-

(1) Carpel - megasporophyll

(2) Stamen - microsporophyll

(3) Pollen sacs - microsporangium

(4) Ovule - microgametophyte

20) Pollen grains of many species cause allergic reactions mainly affecting :-

(1) Digestive system

(2) Neruous system
(3) Respiratory system
(4) Excreatory system

21) Unisexuality of flowers prevents

(1) Geitonogamy, but not xenogamy
(2) Autogamy and geitonogamy
(3) Autogamy, but not geitonogamy
(4) Both geitonogamy and xenogamy

22) In our 60% of angiosperms, pollengrains are shed at .....stage whereas in remaining angiosperms pollen grains are shed at 3 celled stage.

(1) 3-celled
(2) 2-celled
(3) 5-celled
(4) 1-celled

23) Identify incorrect statement from the options given below :

(1) Mesophyll cells in the leaves, have a large number of chloroplasts.
(2) Chloroplast align themselves along the walls of the mesophyll cells according to light intensity
(3) There is no division of labour within the chloroplast.
(4) Membrane system of chloroplast is responsible for trapping the light energy.

24) What are the products of light reactions ?

(1) ATP and NADPH
(2) Sugar
(3) CO_2
(4) All of the above

25) Which of the following factor is major limiting for photosynthesis ?

(1) Light
(2) CO_2
(3) Temperature
(4) O_2

26) Which of the following is first stable product in C4 cycle ?

(1) Oxaloacetic acid
(2) RuBP
(3) Phosphoenol pyruvic acid
(4) 3-phosphoglyceric acid

27) How many ATP will be formed by partial oxidation of Glucose in absence of oxygen.

(1) 2 ATP
(2) 38 ATP
(3) 8 ATP
(4) 30 ATP

28) Electron transport system and ATP synthase are present in which of the following parts of mitochondria ?

(1) Outer membrane
(2) Inner membrane
(3) Matrix
(4) Peri mitochondrial space

29) Which of the following is the first complex of ETS ?

(1) ATP synthase
(2) NADH dehydrogenase
(3) Succinate dehydrogenase
(4) Cytochrome bc_1

30) Filling the blank
All carbohydrates are usually first converted into _____ before they are used for respiration.

(1) Glucose
(2) Sucrose
(3) Starch
(4) Lipid

31) The differences in the shape of leaves produced in air and water is represented by:

(1) Cotton
(2) Coriander
(3) Larkspur
(4) Buttercup

32) E. Kurosawa reported symptoms of foolish seedling disease in :–

(1) Canary grass
(2) Rice
(3) Oat
(4) Orange

33) Ethylene is used to initiate flowering and for synchronising fruit set in –

(1) Rice

(2) Potato

(3) Pineapples

(4) Cherry

34) Geometric growth is related to -

(1) $W_o = W_1 e^{rt}$

(2) $W_o = W_1 + e^{rt}$

(3) $W_1 = W_o e^{rt}$

(4) $W_1 = W_o + e^{rt}$

35) How many of the statements are correct about ethylene ?
(i) It promotes rapid internode elongation in deep water rice plant.
(ii) It may accelerates abscission in flowers and fruits (thinning of cotton, cherry, walnut).
(iii) It has ability to cause an increase in length of axis thats why used to increase the length of grapes stalks and in increase the length of sugarcane stem.

(1) i, ii, iii

(2) iii, ii

(3) i, ii

(4) i, iii

36) Given below two statements.
Statement-I : Heterophylly in cotton, coriander and larkspur are examples of plasticity.
Statement-II : Gibberellins, cause fruits like apple to elongate and improve its shape.
Choose the correct answer from the options given below.

(1) Both **Statement I** and **Statement II** are correct.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are incorrect.

37) Read the following terminology of column-I & II and identify their correct match :

	Column-I		**Column-II**
(A)	Abscisic acid	(i)	Promotes female flower in cucumber
(B)	Auxin	(ii)	Promotes bolting in beet & cabbage
(C)	Cytokinin	(iii)	Induce parthenocarpy in tomato
(D)	Gibberelins	(iv)	Stimulates the closure of stomata
		(v)	Promotes root growth & root hair formation
		(vi)	Promotes nutrient mobilisation
		(vii)	Break seed & bud dormancy

(1) A-iii, B-v, C-vi, D-ii

(2) A-iv, B-iii, C-v, D-ii

(3) A-iv, B-iii, C-vi, D-ii

(4) A-i, B-iv, C-iii, D-v

38) **Assertion (A) :-** When protein are respiratory substrates the ratio would be about 0.9.
Reason (R) :- Pure proteins or fats are never used as respiratory substrates.
In the light of the above statement, choose the most appropriate answer from the options given below :

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

39) According to Amphibolic pathway
Choose incorrect statement –

(1) Fats would had to be broken down into glycerol and fatty acid first

(2) Glycerol would enter the pathway after being converted to PGAL

(3) Proteins would be degraded by the lipase.

(4) All of the above

40) Match the column I with column II

	Column I		**Column II**
i	Pyruvic acid → Acetyl CoA	a	Glycolysis
ii	Glucose → Glucose 6 phosphate	b	Fermentation
iii	Citric acid → α-ketoglutaric acid	c	Link Reaction
iv	Pyruvic acid → Lactic acid	d	Kreb's cycle

(1) i - c, ii - a, iii - d, iv - b

(2) i - d, ii - c, iii - a, iv - b

(3) i - c, ii - d, iii - b, iv - a

(4) i - a, ii - b, iii - c, iv - d

41) How many of the following statement is/are correct.
(a) Calculations of net gain of ATP is only a theoretical exercise.
(b) There is always sequential orderly pathways functioning in respiration.
(c) The NADH synthesised in glycolysis always transferred into mitochondria.
(d) None of the intermediates in the Respiratory pathway are utilised to synthesise any other compound.

(1) Four

(2) Three

(3) Two

(4) One

42) Choose the incorrect statement from the following :-

(1) The protons that are produced by the splitting of water accumulate within the lumen of the thylakoids.

(2) As electrons move through the photosystem, protons are transported across the membrane.

(3) ATP synthase has a channel that allows diffusion of protons back across the membrane.

(4) ATP synthase consists of two part one called CF_0 protrudes on the outer surface of the thylakoid membrane and other portion CF_1 embedded in the thylakoid membrane.

43) Match **List I** with **List II** :

	List–I		**List–II**
A.	Joseph Priestley	I	Discoverer of bacterial photosynthesis
B.	Jan ingenhousz	II	Action spectrum of photosynthesis
C	T.W. Engelmann	III	Green part of the plant could release O_2
D	C.V. Neil	IV	Discoverer of O_2

Choose the correct answer from the options given below.

(1) A–I, B–II, C–III, D–IV

(2) A–III, B–IV, C–II, D–I

(3) A–IV, B–III, C–II, D–I

(4) A–IV, B–III, C–I, D–II

44)

What happens during photorespiration?

(1) Neither synthesis of sugars, nor of ATP

(2) Synthesis of both sugar and ATP

(3) Synthesis of sugar only

(4) Synthesis of ATP only

45) Identify the correct statements.

A. In C_3 plants first CO_2 acceptor was a five carbons ketose sugar.

B. Plants that can be adapted to dry tropical regions may have the C_4 pathway.

C. C_4 Plants shows kranz anatomy in their leaves.

D. In photorespiration RuBisCO shows carboxylase activity.

Choose the correct answer from the options given belove

(1) A and D only

(2) A, B and D only

(3) B and C only

(4) A, B and C only

46) Match the following column-I with column-II and choose the correct option:-

Column-I		Column-II	
A	Apolar neuron	i	Retina of Eye
B	Multipolar neuron	ii	embryonic stage
C	Unipolar neuron	iii	Hydra
D	Bipolar neuron	iv	Cerebral cortex

(1) A-iii, B-iv, C-ii, D-i

(2) A-ii, B-iii, C-i, D-iv

(3) A-iii, B-ii, C-iv, D-i

(4) A-iii, B-iv, C-i, D-ii

47) _____ is a part of peripheral nervous system that comprises the whole complex of nerves, ganglia, nerve fibres and plexuses.

(1) Sympathetic neural system

(2) Parasympathetic neural system

(3) Visceral nervous system

(4) Somatic neural system

48) Given below are two statements.
Statement-I: A synapse is formed by the membranes of pre-synaptic neuron and post synaptic neuron. Which is always separated by a gap called synaptic cleft.
Statement-II: Transmission of an impulse across electric synapse is very similar to impulse conduction along a single axon.

(1) Both statement-I and statement-II are correct`

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct.

49) Identify the statement that is NOT correct.

(1) Fibres of the tracts are covered with the myelin sheath, which constitute the outer part of cerebral hemisphere.

(2) The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocacupus etc form a complex structure called limbic system.

(3) The cerebrum wraps around a structure called thalamus.

(4) The hypothalamus contains a number of centres which control body temperature, urge for eating and drinking.

50) The nervous system is a highly specialized system that control and co-ordinates body activities. It works by transmitting electrical impulses through neurons. The functional unit of the nervous system is the neuron, which consists of dendrites, cell body and axon. The speed of impulse conduction depends on presence or absence of myelin sheath.
Statement-I :- Myelinated nerve fibres conduct nerve impulses faster than non-myelinated nerve

fibres.
Statement-II :- Myelin sheath enables saltatory conduction of nerve impulses from one node of Ranvier to the next.
Given above statements choose the most appropriate answer :

(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

51) A flow chart shows:
Stimulus → Na^+ channels open → Depolarisation → K^+ channels open → Resting potential
Which of following step ensures repolarisation?

(1) Na^+ influx
(2) Ca^{2+} influx
(3) K^+ efflux
(4) Cl^- influx

52) Amygdala and Hippocampus are the parts of–

(1) Fore brain
(2) Mid brain
(3) Hind brain
(4) Brain stem

53) Which part of neuron transmits impulses towards cell body ?

(1) Dendron
(2) Axon
(3) Telodendria
(4) Myelin sheath

54) Excess secretion of growth hormones in adults lead to:-

(1) Gigantism
(2) Ateliosis
(3) Acromegaly
(4) Cretinism

55) PTH stimulates reabsorption of Ca^{++} from the:-

(1) Digested food
(2) Liver
(3) Blood
(4) Renal tubules

56) Which of the following set of hormone is not antagonestic to each other ?

(1) Insulin-glucagon

(2) Somatocrinin-somatostatin

(3) TCT-PTH

(4) FSH-ADH

57) Match the following column-I with column-II and choose the correct answer.

Column-I		**Column-II**	
(A)	Corticoids	i	Diabetes Mellitus
(B)	Insulin	ii	Grave's disease
(C)	Thyroid hormone	iii	Acromegaly
(D)	Growth hormone	iv	Addison's disease

(1) A-iv, B-i, C-ii, D-iii

(2) A-ii, B-i, C-iii, D-iv

(3) A-iv, B-i, C-iii, D-ii

(4) A-i, B-iv, C-ii, D-iii

58) The corticoids, which are involved in carbohydrate metabolism are called glucocorticoids. Corticoids, which regulate the balance of water an electrolytes in our body are called mineralocorticoids
Given below are two statements about corticoids:
Statement-I: Glucocorticoids stimulate gluconeogenesis, lipogenesis and proteolysis; and inhibit cellular uptake and utilisation of amino acids. Glucocorticoids, particularly cortisol, produces anti-inflammatory reactions.
Statement-II: Mineralocorticoid i.e. aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions.
In the light of the above statements choose the most appropriate answer from the options given below:

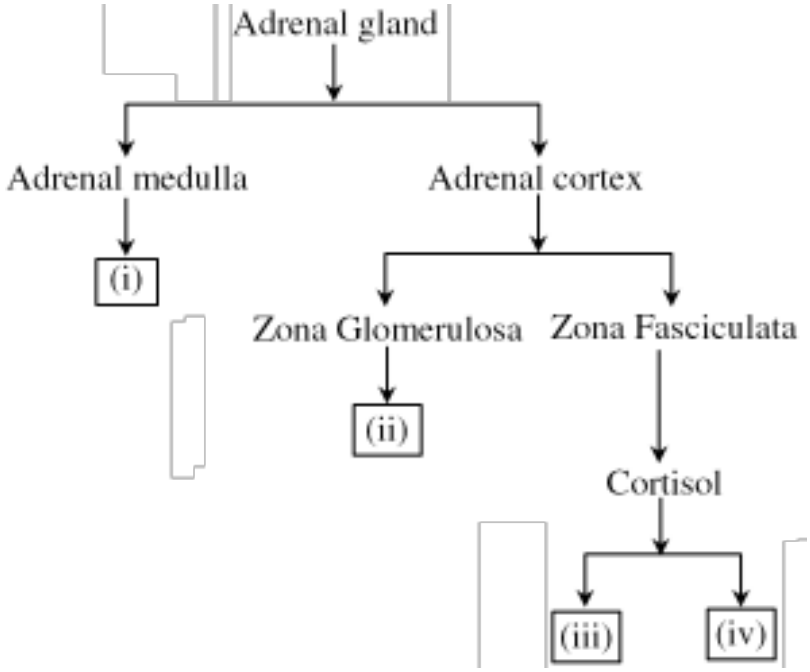
(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II is incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

59) Observe the given classification table. Fill in the boxes (i), (ii), (iii) and (iv) with appropriate



words and select the correct option.

	(i)	(ii)	(iii)	(iv)
1	Catecholamines	Aldosterone	immunosuppresive	erythropoiesis.
2	Catecholamines	Cortisol	immunosuppresive	erythropoiesis
3	Dopamine	Thyroxine	lipolysis	proteolysis
4	Dopamine	Thymosin	Gluconeogenesis	lipolysis

(1) 1

(2) 2

(3) 3

(4) 4

60) **Statement I :** Insulin stumulates glycogensis while glucagon stimulates glyconeogensis
Statement II : Hypothyrodisim during pregnancy causes defective development and maturation of the growing baby.

(1) Both Statements are incorrect

(2) Both Statements are correct

(3) Only Statements I is correct

(4) Only Statement II is correct

61) Select the incorrect statement for the thyroid gland –

(1) Deficiency of iodine in our diet results in hypothyroidism.

(2) Thyrocalcitonin hormone increases the Ca^{2+} levels in the blood.

(3) The thyroid gland is composed of follicles and stromal tissues.

(4) Hyperthyroidism during pregnancy causes cretinism.

62) The α-cells of pancrease secrete hormone called ____.

(1) Insulin
(2) Somatostatin
(3) Pancreatic polypeptide
(4) Glucagon

63) Disruption of the normal sleep wake cycle due to work at night primarily affects which endocrine gland ?

(1) Thymus
(2) Pineal
(3) Anterior pituitary
(4) Parathyroid

64) Which of the following hormones is responsible for contraction of smooth muscle of uterus ?

(1) Prolactin
(2) Oxytocin
(3) Progesterone
(4) Relaxin

65) Which of following is a pair of unpaired bones ?

(1) Frontal, Parietal bones
(2) Parietal, Temporal bones
(3) Nasal, Zygomatic bones
(4) Frontal, Occipital bones

66) Which type of joint occurss betwen carpal and metacarpal to thumb ?

(1) Hinge joint
(2) Pivot joint
(3) Gliding joint
(4) Saddle joint

67) Which of the following bones are not included in the axial skeleton ?

(1) Cranial bones
(2) Ribs
(3) Girdle bones
(4) Facial bones

68) Which type of arrangement is found in cardiac muscle cells ?

(1) Circular pattern
(2) Branching pattern
(3) Linear pattern

(4) Random pattern

69) **Assertion (A):** Cross-bridge formation and breakage occur repeatedly during muscle contraction as long as calcium ions remain in the sarcoplasm.
Reason (R): Calcium ions bind with troponin, removing the masking of actin's active sites and allowing myosin binding.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

70) **Statement I :** Many monomeric proteins called meromyosin constitute one thin filament.
Statement II : Each meromyosin has two important parts a globular head with a short arm and a tail.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

71) Name the molecule responsible for the detachment of myosin from actin during mechanism of muscle contraction.

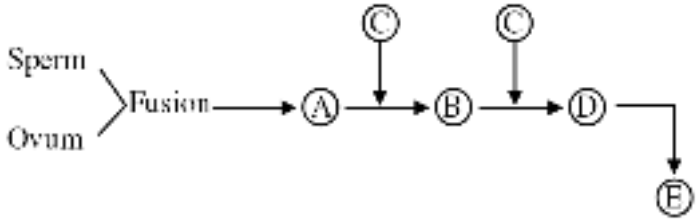
(1) ADP

(2) ATP

(3) Creatine phosphate

(4) GTP

72) Following are the events related to human reproduction



Find the option which correctly identify the A, B, C, D, E

	A	B	C	D	E
(1)	Zygote	Morula	Blastocyst	Cleavage	Gastrula
(2)	Zygote	Embryo with blastomeres	Cleavage	Morula	Blastocyst
(3)	Zygote	Embryo with blastomeres	Morula	Blastocyst	Gastrula
(4)	Zygote	Embryo with blastomeres	Cleavage	Blastocyst	Morula

(1) 1

(2) 2

(3) 3

(4) 4

73) Select the correct option which includes only haploid cells.

(1) Spermatogonia, First polar body, Spermatozoa, Spermatid

(2) Oogonia, Spermatid, Sertoli cells, Granulosa cells

(3) Spermatid, Spermatozoa, First polar body, Ovum

(4) Ovum, Sertoli cells, Theca cells, Spermatids

74) In the given table fill names of phase Menstrual cycle in the blanks.

Phase of Menses cycle	Another name of phase
(i)	Bleeding phase
(ii)	Proliferative phase
Luteal phase	(iii)

(1) i - Menstrual phase, ii - Oestrogenic phase, iii - Post ovulatory phase

(2) i - Oestrogenic Phase, ii - Bleeding phase, iii - Pre-ovulatory phase

(3) i - Progesterogenic phase, ii - Secretory phase, iii - Bleeding phase

(4) i - Secretory phase, ii - Progesterogenic phase, iii - Bleeding phase

75) Match Column-I with Column-II

	Column-I		**Column-II**
(A)	Cortisol	I	Placenta
(B)	HCG	II	Posterior Pituitary
(C)	Oxytocin	III	Sertoli cell
(D)	Androgen	IV	Adrenal cortex
		V	Anterior Pituitary
		VI	Leydig cell

(1) A-V, B-I, C-II, D-III

(2) A-IV, B-V, C-VI, D-II

(3) A-IV, B-I, C-II, D-VI

(4) A-III, B-I, C-II, D-IV

76) A female is unable to give birth to the child due to insufficient secretion of parturating hormone. The doctor decided to administer an injection to stimulate uterine constriction and facilitate normal delivery. Which hormone will be injected ?

(1) Progesteone

(2) Oestrogen

(3) LH

(4) Oxytocin

77) During follicular phase, the primary follicles in the ovary grow to become a fully mature graafian follicle because of

(1) Increase secretion of FSH and LH

(2) Decrease secretion of FSH and LH

(3) Increase secretion of Progesterone

(4) Decrease secretion of Estrogen

78) Match the List-I with List-II and choose the correct option.

	List-I		**List-II**
(A)	Sertoli cells	(i)	Unpaired
(B)	Prostate gland	(ii)	Paired
(C)	Labia minora	(iii)	Seminiferons tubule
(D)	Mammary glands	(iv)	Female secondary sexual character

(1) A-ii ; B-iii ; C-i ; D-iv

(2) A-iv ; B-iii ; C-ii ; D-i

(3) A-i ; B-iii ; C-iv ; D-ii

(4) A-iii ; B-i ; C-ii ; D-iv

79) Which of the following cells in the seminiferous tubules provide nutrition to the dividing germ cells?

(1) Leydig cells

(2) Sertoli cells

(3) Spermatogonia

(4) Follicular cells

80) During folliculogenesis, the secondary follicle transforms into tertiary follicle which is characterised by-

(1) Theca layer

(2) Zona pellucida

(3) Antrum

(4) Corona radiata

81) Sperm are temporarily stored and matured in :-

(1) Prostate

(2) Epididymis
(3) Ampulla
(4) Rete testis

82) Relaxin is secreted by –

(1) Testis
(2) Hypothalamus
(3) Anterior pituitary
(4) Ovary

83) Which of the following is correct regarding menstrual cycle?

(1) Rapid secretion of LH leads to its maximum level at the end of cycle.
(2) Corpus luteum secretes large amount of progesterone.
(3) LH surge induces development of follicles.
(4) Corpus luteum degenerates after fertilization.

84) **Assertion (A) :** The first sign of growing foetus may be noticed by listening to the heart sound carefully through the stethoscope.
Reason (R) : During one month of pregnancy, the embryo's heart is formed.

(1) A and R both are correct and R explains assertion correctly.
(2) A and R both are correct but and R do not explains assertion correctly.
(3) A is correct and R is incorrect.
(4) A is incorrect and R is correct.

85) Match the following columns and select the correct option

	Column-I		**Column-II**
(A)	Barrier Methods	(I)	Diaphragm
(B)	IUD	(II)	LNG-20
(C)	Surgical methods	(III)	Vasectomy
		(IV)	Tubectomy
		(V)	Vault
		(VI)	Condom
		(VII)	Femidom
		(VIII)	Cervical caps

(1) (A)- I, V, VI; (B)-II, III, IV; (C)-VII, VIII
(2) (A)- I, II, V, VI; (B)-VIII, II; (C)-III, IV
(3) (A)- I, V, VI, III; (B)-VII, II; (C)-III, V
(4) (A)- I, V, VI, VII, VIII; (B)-II; (C)-III, IV

86) The table shows classification of contraceptive methods.

Category	Example
Natural	Periodic abstinence
(i)	Cu-T
Barrier Method	(ii)
(iii)	Tubectomy

(1) i-Hormonal, ii-Chemical, iii-injectible

(2) i-IUD, ii-condom, iii-surgical

(3) i-IUD, ii-Hormonal, iii-surgical

(4) i-Hormonal, ii-condom, iii-injectible

87) 

Identify above figure :-

(1) Condom

(2) Implant

(3) CuT

(4) LNG-20

88) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): Oral administration of small doses of either progestogens or progestogen–estrogen combinations is another contraceptive method used by the females.
Reason (R): Oral pills inhibit ovulation (by decreasing the secretion of FSH and LH) and implantation as well as alter the quality of cervical mucus to prevent/retard entry of sperms.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

89) Which of the following contraceptive IUD makes the uterus unsuitable for implantation ?

(1) Cu-T

(2) Cu-7

(3) LNG-20

(4) Multiload-375

90) Given below are two statements: **Statement I :** The term 'reproductive health' simply refers to

healthy reproductive organs with normal functions.
Statement II : Amniocentesis is used to test for the presence of STDs.
Choose the correct answer from the options given below.

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** & **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	3	2	1	1	3	2	4	4	2	3	4	4	4	3	2	2	1	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	3	1	3	1	4	2	1	1	4	4	3	2	1	2	1	1	3	3
Q.	41	42	43	44	45															
A.	3	4	3	1	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	1	3	3	1	2	4	2	3	3	3	3	2	2	2	1	2	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	1	1	4	4	1	1	3	2	3	3	4	2	3	2	1	1	4	2
Q.	86	87	88	89	90															
A.	2	1	1	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	2	1	2	2	1	3	3	3	1	4	2	2	3	4	4	3	4	4	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	3	1	2	1	1	2	2	1	4	2	3	3	3	1	3	2	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	3	1	4	1	3	4	1	1	3	1	1	3	4	4	1	4	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	2	2	4	4	3	2	1	4	2	2	3	1	3	4	1	4	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	4	2	1	4	2	2	1	3	3										

SOLUTIONS

PHYSICS

1)

$Kx = m_1g \Rightarrow x = \frac{m_1g}{K}$

2) Restoring force and displacement are always opposite to each other.

3) $T = 2\pi\sqrt{\frac{m}{K}}$
which can be changed by (A), (C)

4) NCERT Pg. # 268
$K_1 = K_2 = 2K = 200$ N/m
$K_{eq} = K_1 + K_2 = 200 + 200 = 400$ N/m
$T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{0.25}{400}} = \frac{2\pi}{40} = \frac{\pi}{20}$ sec.

5) The velocity of particle executing S.H.M at its extreme position is zero, hence its momentum is also zero.

6) $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$
If radius is doubled and length is doubled, mass per unit length will become four times. Hence
$f' = \frac{1}{2 \times 2\ell}\sqrt{\frac{2T}{4\mu}} = \frac{f}{2\sqrt{2}}$

7)

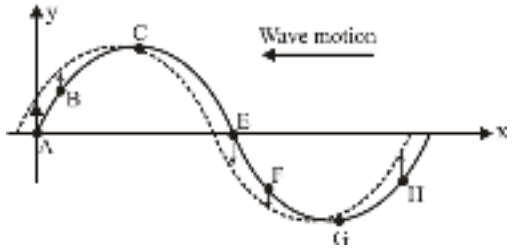
$V = 2f(\square_2 - \square_1)$

$= 2(320) \times (0.73 - 0.20)$

$= (640) \times (0.53) = 339.2$ m/s

8) $v = \frac{\omega}{k} = \frac{0.50}{0.02} = 25$
$v = \sqrt{\frac{T}{\mu}}$ or T = 25N

9) $v_{wave} = \frac{\omega}{k}$
$v_{particle} = A\omega$
On comparing
$v_{w1} = 2m/s$ $v_{w2} = 3$ m/s $v_{w3} = \frac{11}{5}m/s$
$v_{p1} = 12$ cm/s $v_{p2} = 36$ cm/s $v_{p3} = 44cm/s$



10)
Since the wave is moving from right to left, you can imagine shifting the whole wave shape slightly to the left to see where the particles will be in the next moment.
• The part of the wave just to the right of D is lower, so D must move down.
• The part to the right of E is also lower, so E moves down.
• The part to the right of F is the trough, which is lower, so F must also move down to meet it.

Final Answer:(2)

11)

$f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$
$f_1 = \frac{1}{2}\sqrt{\frac{6}{\mu}}$ $f_2 = \frac{1}{2}\sqrt{\frac{54}{\mu}}$
$\frac{f_1}{f_2} = \frac{1}{3}$ $f_2 - f_1 = 12$
$f_1 = 6HZ$

12) Heat lost = Heat gain
$100 \times 0.1(T - 75) = 170 \times 1 \times (75 - 30) + 100 \times (75 - 30)$
$10(T - 75) = (170 + 10)(75 - 30)$
$10(T - 75) = 180 \times 45$
$T = 885°C$

13) (ii) is true according to Boyle's law and (iv) is true according to Charle's law (i.e., constant pressure law)

14) $H = 200 \times 10^{-3} \times 1 \times 20 = 4$ Kcal.
Heat supplied = 4000 cal

15) Equation of PV graph $P = -\left(\frac{P_0}{V_0}\right) V + 3P^0$
$= mV + C$ [$m = -P_0/V_0$ & $C = 3P_0$]
putting this in ideal gas equation $PV = nRT$

We get, $mV^2 + CV = RT$; (n = 1) differentiating $\Rightarrow 2mV + C = R\frac{dT}{dV}$
For T_{max}, $\frac{dT}{dV} = 0 \Rightarrow V = -\frac{C}{2m} = \frac{3V_0}{2}$

16) $P = \frac{dQ}{dt} = kA\frac{\Delta T}{\ell}$
$1000 = \frac{0.2(5)(T-25)}{4\times10^{-2}}$
$T-25 = \frac{4\times10}{0.2\times5} = 40$
T = 65°C

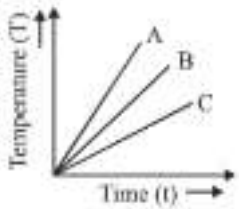
17)

By energy conservation, $E_1 + E_2 = E_{mix}$
$\Rightarrow \frac{n_1 f_1 RT_1}{2} + \frac{n_2 f_2 RT_2}{2} = \frac{(n_1+n_2) f RT_{mix}}{2}$
$\Rightarrow \frac{3}{2}P_1V_1 + \frac{3}{2}P_2V_2 = \frac{3}{2}P_{mix}(V_1+V_2)$
$\Rightarrow \frac{3}{2}\times1\times2 + \frac{3}{2}\times2\times3 = \frac{3}{2}\times P_{mix}(5)$
$\therefore P_{mix} = \frac{8}{5} = 1.6$ atm.

18) PV^γ = constant & $\frac{C_P}{C_V} = \gamma = \frac{3}{2} = 1.5$
$\left(\frac{\Delta P}{P}\times100\right) + \gamma\left(\frac{\Delta V}{V}\times100\right) = 0$
$\Rightarrow \left(\frac{\Delta V}{V}\times100\right) = -\left(\frac{\Delta P}{P}\times100\right)\times\frac{1}{\gamma}$
$\Rightarrow \left(\frac{\Delta V}{V}\times100\right) = -6\%\times\frac{2}{3}$
$= -4\%$

19)

For constant heat flow rate $\frac{dQ}{dt} = ms\frac{dT}{dt}$ = Constant
slope= $\frac{dT}{dt} \propto \frac{1}{s}$



From the graph:
Slope of A > Slope of B > Slope of C $\Rightarrow s_A < s_B < s_C$

20) According to Stefan-Boltzmann law, the energy radiated per second through the surface of area A is given by
$E = \sigma AT^4$

$$\frac{E_1}{E_2} = \frac{A_1}{A_2}\left(\frac{T_1}{T_2}\right)^4$$

$$\text{or } 10000 = \frac{r_1^2}{r_2^2}\left(\frac{2000}{6000}\right)^4$$

$$\text{or, } \frac{r_1^2}{r_2^2} = (30)^4$$

or $r_1 : r_2 = 900 : 1$

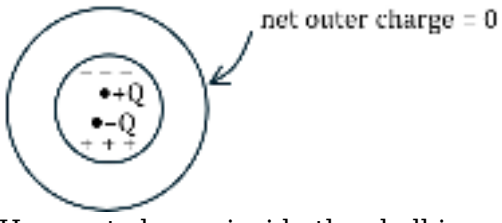
21) $V_{O_2} = \frac{1}{3}V_{H_2}$

$$\sqrt{\frac{273+t}{32}} = \frac{1}{3}\sqrt{\frac{273-3}{2}}$$

$$\text{or } \frac{273+t}{32} = \frac{1}{9} \times \frac{270}{2}$$

$$\text{or } 273 + t = \frac{270 \times 32}{18}$$ or $t = 480 - 273 = 207°C$

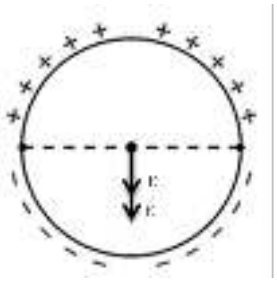
22)



Here net charge inside the shell is zero.
Thus, induced charge on inner surface of shell is zero.
So, induced charge on outer surface of shell will be zero.

$$\sigma = \frac{q}{\text{Area}} = 0$$

Note : (net internal charge will also be zero, average internal charge density will also be zero, but on visiting internal surface there will be respective positive and negative surface parts)
Final Answer: option (3)



23)

$E_{net} = 2E$

Potential at the centre is zero (Total charge is zero)

Assertion (A) is true and Reason (R) is false.

24)



(q_1 and q_2 are positive)

Direction of force on q_2 is along the line joining q_1 to q_2 (away if both are positive)

Unit vector from q_1 to q_2

$$\hat{r}_{12} = \frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}$$ (r_{12} = unit vector from q_1 to q_2)

$\vec{r}_1 = (-1,\ 1,\ 1),\ \vec{r}_2 = (3,\ 1,\ -2)$

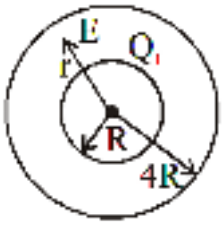
$\vec{r}_2 - \vec{r}_1 = (3-(-1)\hat{i} + (1-1)\hat{j} + (-2-1)\hat{k}$

$= (4\hat{i} + 0\hat{j} - 3\hat{k})$

$|\vec{r}_2 - \vec{r}_1| = \sqrt{4^2 + 0^2(-3)^2} = 5$

$\hat{r}_{12} = \frac{4\hat{i} - 3\hat{k}}{5}$

25) Work done on moving a charge on equipotential surface is zero.



26)

$$E = \frac{KQ_1}{r^2}$$

$$\Delta V = \int_R^{4R} E\ dr = \frac{3KQ_1}{4R}$$

27) As $\vec{E}$ is a vector quantity.

28) $\phi_{net} = 0$ as $\phi = \phi_{entering} - \phi_{leaving}$

and $\phi_{entering} = \phi_{leaving} = \frac{q}{6\varepsilon_0}$

29) We have centripetal force equation $q\left(\frac{2k\lambda}{r}\right) = \frac{mv^2}{r}$

so $v = \sqrt{\frac{2kq\lambda}{m}}$

Now, $T = \frac{2\pi r}{v} = 2\pi r\sqrt{\frac{m}{2k\lambda q}}$ where $k = \frac{1}{4\pi\varepsilon_0}$

30) Assertion is true because when a capacitor in connected to a battery the battery puzser charge onto one plate (positive plate) and takes an equal amount of charge form the other plate (negative plate) The charger on both plates will be equal in magnitude, regardless the size of the plate. Reason is also true, the conservation of charge principle ensure that the charge remains constant.

31)

Initially $Q = \frac{CE}{2}$

After inserting slab

$Q' = \frac{KCC}{KC + C} = \frac{KCE}{K+1}$

$Q' - Q = \frac{KCE}{K+1} - \frac{CE}{2}$

Charge supplied by battery = $\frac{(K-1)CE}{2(K+1)}$

32) Before closeing the switch capacitor is fully charged. So, no current flows in the circuit. After switch is closed no current will flow because it will have infinite resistance. Hence heat produce is zero.

33)

After redistribution new charges on spheres are

$Q_1' = \left(\frac{10}{10+20}\right) \times 10 = \frac{10}{3}\mu C$

and $Q_2' = \left(\frac{20}{10+20}\right) \times 10 = \frac{20}{3}\mu C$

Ratio of charge densities $\frac{\sigma_1}{\sigma_2} = \frac{Q_1'}{Q_2'} \times \frac{r_2^2}{r_1^2}$

$= \frac{10/3}{20/3} \times \left(\frac{20}{10}\right)^2 = \frac{2}{1} \quad \left\{\sigma = \frac{Q}{4\pi r^2}\right\}$

34) Let n be the number of wrongly connected cells.
Number of cells helping one another =(12 – 2n)
Total e.m.f. of such cells =(12 – 2n)E
Total resistance of cells = 12r
(resistance remains same irrespective of connections of cells)
With additional cells

(a) Total e.m.f. of cells when additional cells help battery = (12 – 2n) E + 2E
Total resistance = 12r + 2r = 14r

$$\therefore \quad \frac{(12-2n)E+2E}{14r}=3 \qquad \text{.....(i)}$$

(b) Similarly when additional cells oppose the battery

$$\frac{(12-2n)E-2E}{14r}=2 \qquad \text{.....(ii)}$$

Solving (i) and (ii), n = 1

35)

$R_1 = 3R \qquad R_2 = \frac{2R}{3}$

$R_3 = \frac{R}{3} \qquad R_4 = \frac{3R}{2}$

$P = I^2R \qquad (I \rightarrow \text{same})$

$P \propto R$

$R_3 < R_2 < R_4 < R_1$

$P_3 < P_2 < P_4 < P_1$

36) (b) In series : Potential difference ∝ R

When only S_1 is closed $V_1 = \frac{3}{4}E = 0.75E$

When only S_2 is closed $V_2 = \frac{6}{7}E = 0.86E$

and when both S_1 and S_2 are closed combined resistance of 6R and 3R is 2R

$$\therefore \quad V_3 = \left(\frac{2}{3}\right)E = 0.67E \qquad \Rightarrow V_2 > V_1 > V_3$$

37) $R_1' = R_1(1+\alpha_1\theta);\ R_2' = R_2(1+\alpha_2\theta)$

$R_{net} = R_1' + R_2' = R_1(1+\alpha_1\theta) + R_2(1+\alpha_2\theta)$

$= R_1 + R_2 + \theta(R_1\alpha_1 + R_2\alpha_2)$

$= (R_1+R_2)\left[1+\frac{R_1\alpha_1+R_2\alpha_2}{R_1+R_2}\theta\right]$

Compair this equation $R = R_0(1+\alpha_{net}\theta)$

$\therefore \quad \alpha_{net} = \frac{R_1\alpha_1 + R_2\alpha_2}{R_1+R_2}$

If $R_1 = R_2 = R_0$ Then

$\alpha_{net} = \frac{\alpha_1+\alpha_2}{2}$

38) Applying Kirchhoff's first law at the junction P, we get : $6 = I_1 + I_2$ .....(i)

Applying Kirchhoff's second law to the closed circuit PQRP, we get :

$-2I_1 - 2I_1 + 2I_2 = 0$

or $2I_1 + 2I_1 - 2I_2 = 0$

or $4I_1 - 2I_2 = 0$ .....(ii)

Solving eq. (i) and (ii), we get

$I_1 = 2A, I_2 = 4A$

39) $\mu = \frac{V_d}{E} = \frac{7.5 \times 10^{-4}}{3 \times 10^{-10}}$

$\mu = 2.5 \times 10^6$

40) $S = \frac{G}{\frac{I}{I_g} - 1} \Rightarrow S = \frac{45\Omega}{\frac{10A}{1A} - 1} \Rightarrow S = 5\Omega$

41)

Intensity and Amplitude Relationship:

$I \propto A^2 (\omega)^2$(Intensity is proportional to the square of amplitude).

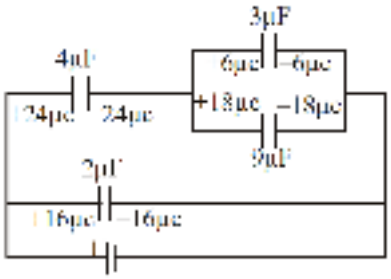
$\frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2 \left(\frac{\omega_1}{\omega_2}\right)^2$

Given that $A_1 = 4\ A_2$ and $\omega_2 = 5\ \omega_1$

then Intensity Ratio :

$\frac{I_1}{I_2} = \left(\frac{4}{5}\right)^2 = \frac{16}{25}$

Final Answer : Option (3)



42) $Q = 24 + 18 = 42\ \mu c$

$E = \frac{KQ}{r^2} = \frac{9 \times 10^9 \times 42 \times 10^{-6}}{30^2} = 420\ N/C$

43) Given circuit is balanced wheat stone bridge.

Hence bridge branch can be removed.

then R_{eq} = (5 + 7) Parallel with (10 +14) = $\frac{12 \times 24}{12 + 24} = 8\Omega$

44)

$V_{300\Omega} = 60 - 30 = 30V$

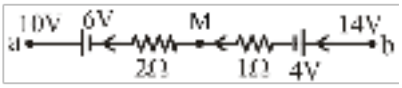
$i = \frac{30}{300} = \frac{1}{10} A = 0.1A$

$i_{400} = \frac{30}{400} = \frac{3}{40} = 0.075A$

$i_v = i - i_{400}$

$$= \frac{1}{10} - \frac{3}{40} = \frac{1}{40}$$

$$R_V = \frac{V}{i_V} = \frac{30}{1/40} = 1200\Omega$$

45) 

Apply KVL from A to B

$V_A - 6 + 2i + i + 4 = V_B$

$10 - 6 + 3i + 4 = 14$

$3i = 6$

$i = 2$ amp

KVL from A to M

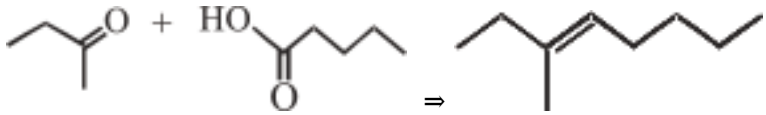
$V_A - 6 + 2i = V_m$

$V_A - 6 + 4 = V_m$

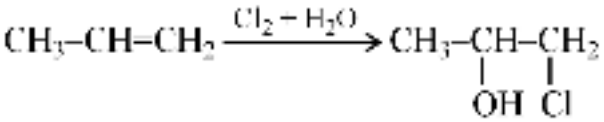
$10 - 2 = V_m$

$V_m = 8$ volt.

CHEMISTRY

46) 

47) Incorrect product of 3rd is



48)

Rate of decarboxylation ∝ Stability of carbanion formed in RDS.

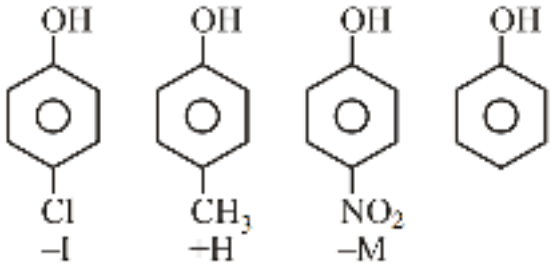
49)

In aqueous phase

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

2° 1° 3°

Question Explanation: Ordering the basic strength of methylamines in an aqueous phase.
Concept: Basicity in Aqueous Medium
Solution: The basicity of methylamines in an aqueous medium is determined by the combined effects of the inductive effect, solvation effect, and steric hindrance. The correct order is (CH3)2NH>CH3NH2>(CH3)3N>NH3. **Calculation**: N/A
Final Answer: 3



(I) (II) (III) (IV)

50)

Acidic strength(A.s/ka)→III>I>IV>II

pK_a → II > IV>I>III

$$\begin{bmatrix} \text{A.S } \alpha - \text{M}, -\text{H}, -\text{I} \\ \alpha \ \ \frac{1}{+\text{M}}, \frac{1}{+\text{H}}, \frac{1}{+\text{I}} \end{bmatrix}$$

51)

Enthalpy of Hydrogenation $\propto \frac{1}{\text{Stability of alkene}}$

52)

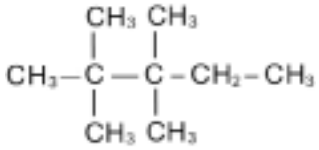
Theoretical

53) **Solution :**

An isopropyl group is a group with the structure $–CH(CH_3)_2$

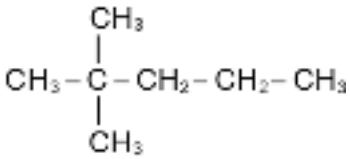
To determine which compound has one isopropyl group. We will analyze each option.

(1) 2, 2, 3, 3-Tetramethylpentane.



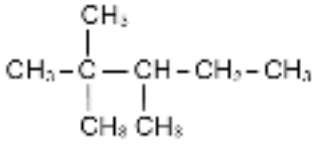
(No isopropyl group)

(2) 2, 2-Dimethylpentane



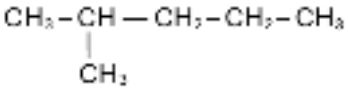
(No isopropyl group)

(3) 2,2,3-Trimethylpentane



(No isopropyl group)

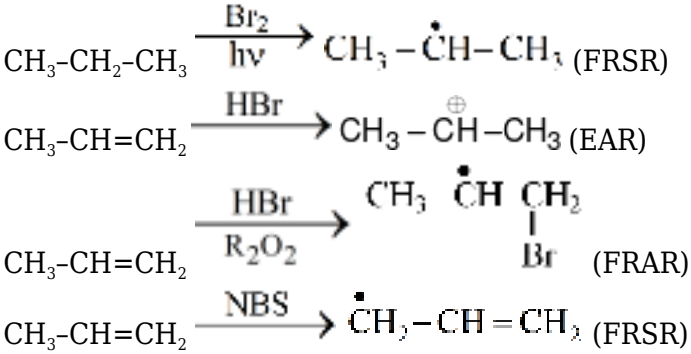
(4) 2-Methylpentane



This compound has (Isopropyl group)

Hence, Option (4) is correct : 2-methyl pentane

54)

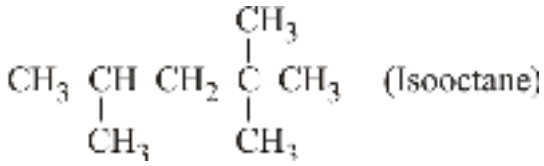


(FRSR)

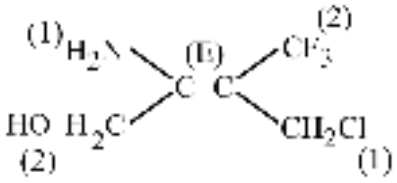
(EAR)

(FRAR)

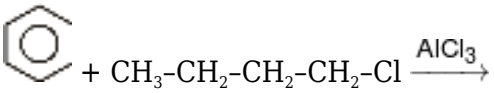
(FRSR)

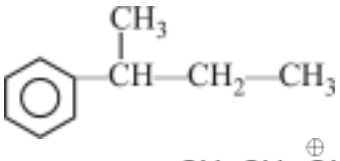
55)  (Isooctane)

56) –F due to H-bonding with –OH group.

57) 

58)

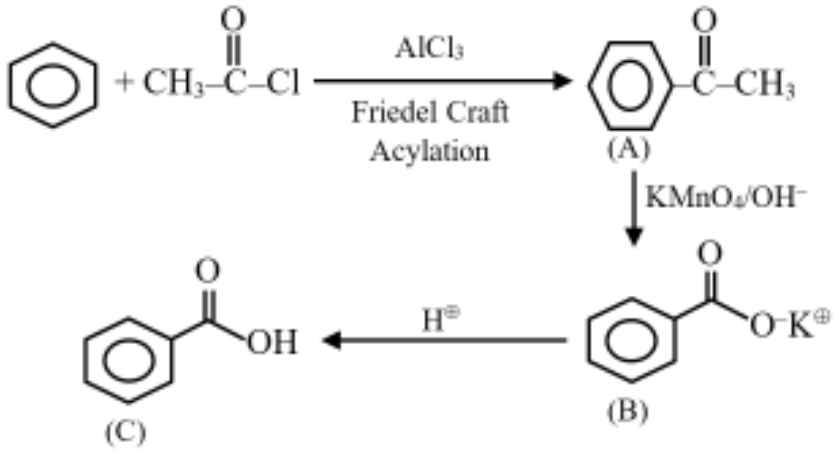




intermidiate $CH_3CH_2\overset{\oplus}{C}H-CH_3$ is formed through $H^{\ominus}$ shifting

Final Answer :- 3

59)



60) $S \propto \frac{-I \text{ effect}}{+I \text{ effect}}$

61)

Stability of $C^{\oplus} \propto \frac{+M, +H, +I}{-M, -H, -I}$

62)

NCERT, Pg.# 285
Present → HNO_3 treatment before $AgNO_3$ test.

63) **Solution:**
1. Under reduced pressure, the vapour pressure of a liquid reaches atmospheric pressure at a lower temperature (Clausius–Clapeyron relation).
2. Liquids with high normal boiling points often decompose before reaching those temperatures.
3. By distilling under vacuum, their boiling points drop, allowing gentle purification without thermal degradation. Hence, vacuum (reduced-pressure) distillation is especially used for liquids that have high boiling points.
Final answer: Option (2)

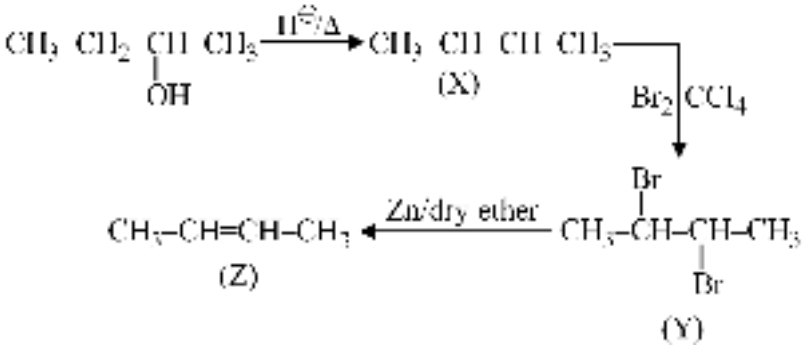
64)

Alkane having 3°H can react with $KMnO_4$ to give alcohol.

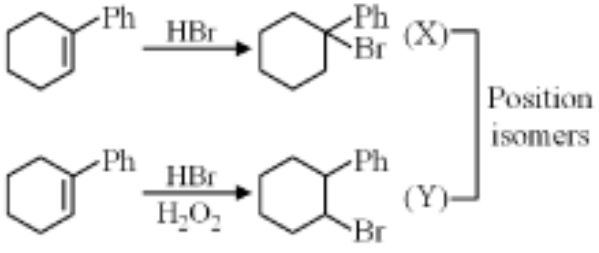
65)

Theoretical

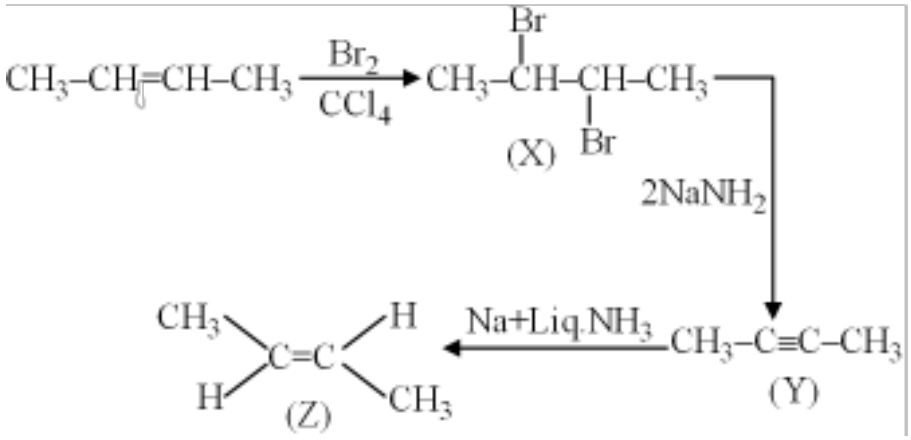
66)



67)



68)



69)

Rate may or may not depends on concentration of reactant.
Catalyst can increase or decrease the ROR.
Rate of reaction depends on temperature.
For solid reactant, Rate depends on surface area at low pressure.

70)

$$Kt = 2.303 \log \frac{C_o}{C_t}$$

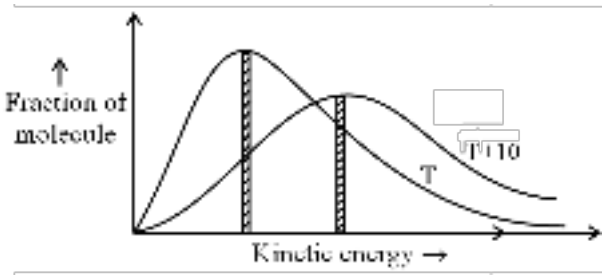
$$30k = 2.303 \log \frac{C_o}{C_1} \quad ...(1)$$

$$40k = 2.303 \log \frac{C_o}{C_2} \quad ...(2)$$

$$\frac{(1)}{(2)} \ \frac{3}{4} = \frac{\log \frac{C_o}{C_1}}{\log \frac{C_o}{C_2}} \Rightarrow 4 \log \frac{C_o}{C_1} = 3 \log \frac{C_o}{C_2}$$

$$\left(\frac{C_o}{C_1}\right)^4 = \left(\frac{C_o}{C_2}\right)^3$$

$$C_o = \frac{C_1^4}{C_2^3}$$

71) 

In MDC, fraction of molecules with most probable kinetic energy decreases and most probable kinetic energy of molecules increases with increases in temperature.

72) (1) on T↑, k↑, so $t_{\frac{1}{2}}$ ↓

(2) for zero order reaction order and molecularity can't be same. So it is always complex reaction.

(3) on T↑, k↑ because $k = Ae^{-\varepsilon_a/RT}$

(4) Activation energy energy does not depend on temp.

73) 36 g $O_2 = \frac{36}{32} = 1.125$ mol/min

$$-\frac{1}{2}\frac{[H_2O_2]}{dt} = \frac{d[O_2]}{dt}$$

$$\therefore -\frac{d[H_2O_2]}{dt} = 2 \times \frac{d[O_2]}{dt} = 2 \times 1.125$$

$= 2.25$ mol/min

74) $$\log\frac{k_2}{k_1} = \frac{\varepsilon_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log 3 = \frac{\varepsilon_{a,A}}{2.303R}\left(\frac{20}{200 \times 220}\right) \quad ...(1)$$

$$2\log 3 = \log 9 - \frac{\varepsilon_{a,B}}{2.303R}\left(\frac{20}{200 \times 220}\right) \quad ...(2)$$

$$\frac{(2)}{(1)} \quad \frac{\varepsilon_{a,B}}{\varepsilon_{a,A}} = \frac{2}{1}$$

$E_{a,B} = 2\ E_{a,A}$

75)

Raoult's law : $p = x_1 p_1^o + x_2 p_2^o$

Henry's law : $p = K_H x$

Elevation of boiling point: $\Delta T_b = K_b m$

Depression in freezing point: $\Delta T_f = K_f m$
Osmotic pressure: π = CST

76) Given curve belong to positive deviation from Raoult's Law. and during this deviation inter molecule attraction force after mixing decreases.

77)

Boiling-point elevation: $\Delta T_b = K_b \times m$.
For 1 m solution: $\Delta T_1 = 0.52 \times 1 = 0.52$ K
For 2 m solution: $\Delta T_2 = 0.52 \times 2 = 1.04$ K
Difference: $T_2 - T_1 = 1.04 - 0.52 = 0.52$ K

78)

When RBC is placed in pure water, it swell up and burst due to endo osmosis.

79)

Using Henry's Law: $P = K_H \times C$ (K_H **of** CO_2
water = 29.76 atm/(mol/L) at 25∘C)
$C = P/K_H = 2.4/29.76 = 0.08$ **mol/L** = 3.52 g/L

80) $\text{For } 1A + 4B : x_A = 1/5,\ x_B = 4/5 \Rightarrow 0.2p_A^0 + 0.8p_B^0 = 560.$ 1
$\text{For } 1A + 3B : x_A = 1/4,\ x_B = 3/4 \Rightarrow 0.25p_A^0 + 0.75p_B^0 = 550.$ 2
Multiply first eqn by 5: $p^0_A + 4\ p^0_B = 2800.$
Multiply second eqn by 4: $p^0_A + 3\ p^0_B = 2200.$
Subtract: $p^0_B = 600$ mm Hg.
Then $p^0_A = 2200 - 3(600) = 400$ mm Hg.

81)

$\Delta T_f = ik_f m$

$$3.72 = i \times 1.86 \times \frac{5 \times 1000}{142 \times 50}$$

i = 2.84

82)

H_2SO_4 : $N_1V_1 = 2 \times 0.1 \times 0.2 = 0.04$ Equivalents
KOH : $N_2V_2 = 0.2 \times 0.15 = 0.03$ Equivalents
H_2SO_4 is in excess 0.03 Equivalents of both H_2SO_4 and KOH neutralize enthalpy of neutralisation = $0.3 \times 57.2 = 1.716$ kJ

83)

(i) $CuSO_4(s) \rightarrow CuSO_4(aq.)$ $\Delta H = -15.89$
(ii) $CuSO_4.5H_2O(s) \rightarrow CuSO_4(aq.)$ $\Delta H = 2.80$
(iii) $CuSO_4(s) \rightarrow CuSO_4.5H_2O(s)$
Eq(iii) = eq(i) – eq(ii)
= –15.89 – 2.80
⇒ –18.69 kcal

84) • **Option 1:** Incorrect; anhydrous and hydrated salts have different lattice and hydration energies, leading to different enthalpies.
• **Option 2:** Incorrect; while many salts increase in solubility, some (like $Ce_2(SO_4)_3$) actually decrease as temperature rises.
• **Option 3:** Incorrect; $KMnO_4$ is a self-indicator, turning the solution pink at the endpoint, so litmus is unnecessary.
• **Option 4:** Correct; it must be standardized against a primary standard like oxalic acid before use in titrations.

85) **Aqueous NaCl:** H_2 at cathode, Cl_2 at anode → (r)
Fused NaCl: $Na^+ \rightarrow Na$ at cathode; Cl_2 at anode → (s)
Dilute H_2SO_4: H_2 at cathode, O_2 at anode → (q)
Concentrated H_2SO_4: H_2 at cathode, peroxodisulphate ($S_2O_8^{2-}$) at anode → (p)

86) $$E_{Mg^{+2}/Mg} = E^o_{Mg^{+2}/Mg} - \frac{0.059}{2}\log\frac{1}{[Mg^{+2}]}$$

$$\underbrace{E_{Mg^{+2}/Mg}}_{y} = \underbrace{E^o_{Mg^{+2}/Mg}}_{C} + \underbrace{\frac{0.059}{2}}_{M}\log\underbrace{[Mg^{+2}]}_{x}$$

M = tan θ = +ve ⇒ θ < 90
C = $E^o_{Mg^{+2}/Mg}$ = –ve

87) $\alpha = \frac{\lambda}{\lambda_\infty} = \frac{20}{400} = \frac{1}{20} = 0.05$
∴ $[H^+] = C\alpha = 1 \times 0.05 = 0.05$
∴ $pH = \log\frac{1}{[H^+]} = \log\frac{1}{0.05}$
$= \log 20 = 1.3$

88)

$$\Lambda_e = \frac{K \times 1000}{N} = \frac{\left(\frac{1}{R}\frac{\ell}{a}\right) \times 1000}{N}$$
$$= \frac{\frac{1}{200} \times 2 \times 1000}{0.1} = \frac{10}{0.1} = 100 \text{ S cm}^2\text{eq}^{-1}$$

89)

$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

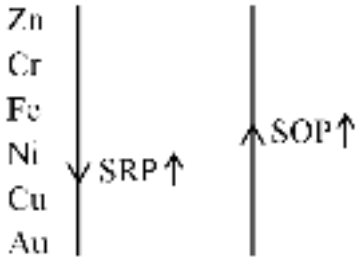
$n_f = 4$

No. of equivalents = $\frac{it}{F}$

$n \times n_f = \frac{i \times t}{F}$

(for 1 seconds) $\frac{2}{22400} \times 4 = \frac{i \times 1}{96500}$

$i = 34.46$ amp/sec.

90) 

Upper metal (Higher SOP) cant be place with lower metal cation (Lower SOP) solution because they both react and metal vessel gets perforated.

Hence $AuCl_3$ solution cannot be stored in vessel made of Cu, Fe, Ni, Cr, Zn

BIOLOGY

91) NCERT-XI, Pg. # 73

92) NCERT-XI, Pg. # 74

93) NCERT-XI, Pg. # 75

94)

Sieve cell present in Gymnosperms.

95)

NCERT, Pg. # 76

96) NCERT, Pg. # 61

97) NCERT (XIth) Pg. # 62, 63

98) NCERT (XIth) Pg. # 64

99)

NCERT, Pg. # 67

100) NCERT-XI, Pg. # 64

101) NCERT-XI, Pg # 59

102)

NCERT-XI, Pg. # 7

103) NCERT Pg. # 25

104) NCERT, Pg # 5

105) NCERT Pg. # 18

106) NCERT Pg. # 12

107) NCERT, Pg. # 4, 5

108) NCERT, Pg. # 7, 8

109) NCERT-XI, Pg. # 9

110)

NCERT-XI, Pg. # 7

111)

NCERT-XI, Pg. # 12

112)

NCERT-XI, Pg. # 7

113)

NCERT-XI, Pg. # 136

114)

NCERT-XI, Pg. # 139, 140

115) NCERT-XI, Pg # 223

116) NCERT-XI, Pg # 216

117) NCERT Pg. No. # 230

118) NCERT-XI, Pg # 232

119) NCERT-XI, Pg # 233

120) NCERT Pg. # 235

121) NCERT XI Pg. # 246

122) NCERT-XI, Pg. # 248

123)

NCERT-XI, Pg. # 177

124) NCERT XI Pg. # 170

125) NCERT-XI, Pg. # 177

126) NCERT XI Pg. # 246, 249

127) NCERT, Pg # 176,177

128) NCERT XI Pg. 236

129) NCERT Pg. # 235

130) NCERT Pg. No. # 229, 230, 231

131) NCERT (XI) Page No. # 234

132) NCERT, Pg. # 213, 214, 215

133)

NCERT XI, Page No. 207, 208

134)

NCERT Pg. # 147

135)

NCERT XI, Page No. 216,218

136) NCERT, Pg. # 232

137) NCERT, Pg. # 231

138) NCERT, Pg. # 234

139) NCERT, Pg. # 236

140) NCERT, Pg # 232

141) NCERT, Pg. # 233

142) NCERT, Pg. # 235

143) NCERT Pg. No. # 232

144) NCERT, Pg. # 241

145) NCERT, Pg. # 243

146) NCERT, Pg. # 241, 243, 244

147) NCERT, Pg. # 241,243,244

148) NCERT, Pg. # 245

149) NCERT, Pg. # 244,245

150) NCERT, Pg. # 242, 245

151) NCERT, Pg. # 242, 243

152) NCERT Pg # 245

153) NCERT Pg # 242

154) NCERT Pg # 241

155) NCERT-XIth, Pg. # 224

156) NCERT Pg. No. # 227

157) NCERT Pg. No. # 224

158) NCERT Pg. No. # 219

159) NCERT, Pg # 222,223

160) NCERT Page No. - 221

161) NCERT XI Pg.#222

162) NCERT pg no # 36

163) NCERT pg no # 32, 33

164) NCERT, Pg. # 34

165) NCERT pg no#37

166) NCERT, Pg. # 38

167) NCERT, Pg. # 34

168) NCERT Pg.# 27, 28, 30, 39

169) NCERT Pg. No. # 39

170) NCERT Pg. # 32

171) Application type - Allen module.

172) NCERT Pg. # 37

173) NCERT Pg. # 34

174) NCERT Pg. # 98

175) NCERT Pg. # 44, 45

176) NCERT Pg. # 44-45

177) NCERT-XII, Pg. No. # 45

178) NCERT, Pg. # 45

179) NCERT Pg. # 44-45

180)

NCERT, Pg. # 41, 42